2025-26 S1 – BIO220 PROGRAM

Week	Lecture	Tutorials and Labs	Coursework
Week 1	Introduction to computational Biology.		
Week 2	Charges and Coulomb law. Electrostatic potential.		
Week 3	Atoms and molecules. The Schrodinger equation		
Week 4	Molecular mechanics. Force Fields.		Coursework 1 (in class test): Molecular Physics.
Week 5	Structure of proteins and nucleic acids.	Tutorial and Lab: Molecular visualization with VMD.	
Week 6	Structural models and protein folding. Homology modeling and Alphafold.	Tutorial and Lab: Homology modeling.	Coursework 2 (laboratory): Homology modeling and molecular visualization.
Week 7	Molecular surfaces and electrostatics.		
Week 8	Molecular docking.		Coursework 2 deadline
Week 9	Energy minimization. Normal modes analysis.	Tutorial and Lab: Molecular docking, normal modes.	
Week 10	Free energy and solvation.		
Week 11	Small molecules and chemical spaces. Chemoinformatics.	Tutorial and lab: Protein/Ligand modeling.	Coursework 3 (laboratory): Protein/Ligand modeling.
Week 12	Drug libraries, screening and selection.		
Week 13	Recap.		Coursework 3 deadline

PROTEIN FOLDING

Francesco Zonta, Ph.D.

Associate Professor

Department of Biosciences and Bioinformatics

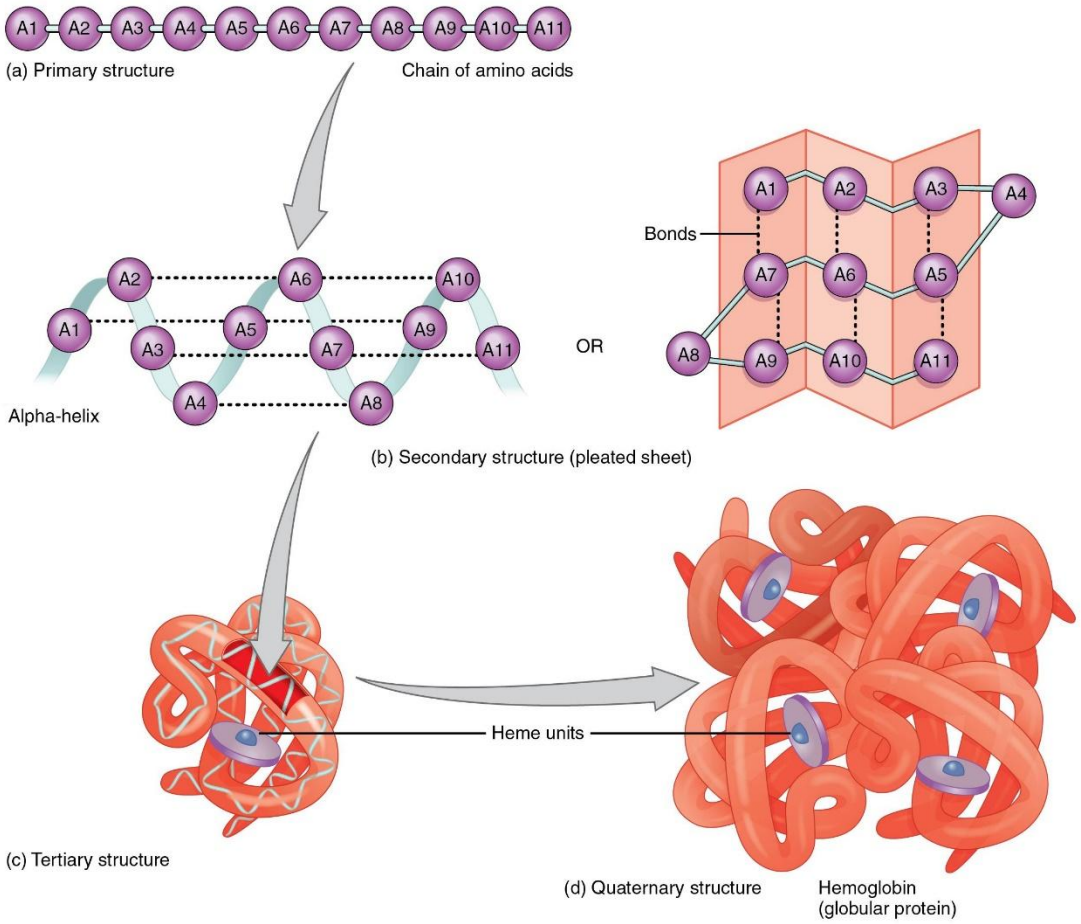
Room SB465, francesco.zonta@xjtlu.edu.cn



Xi'an Jiaotong-Liverpool University

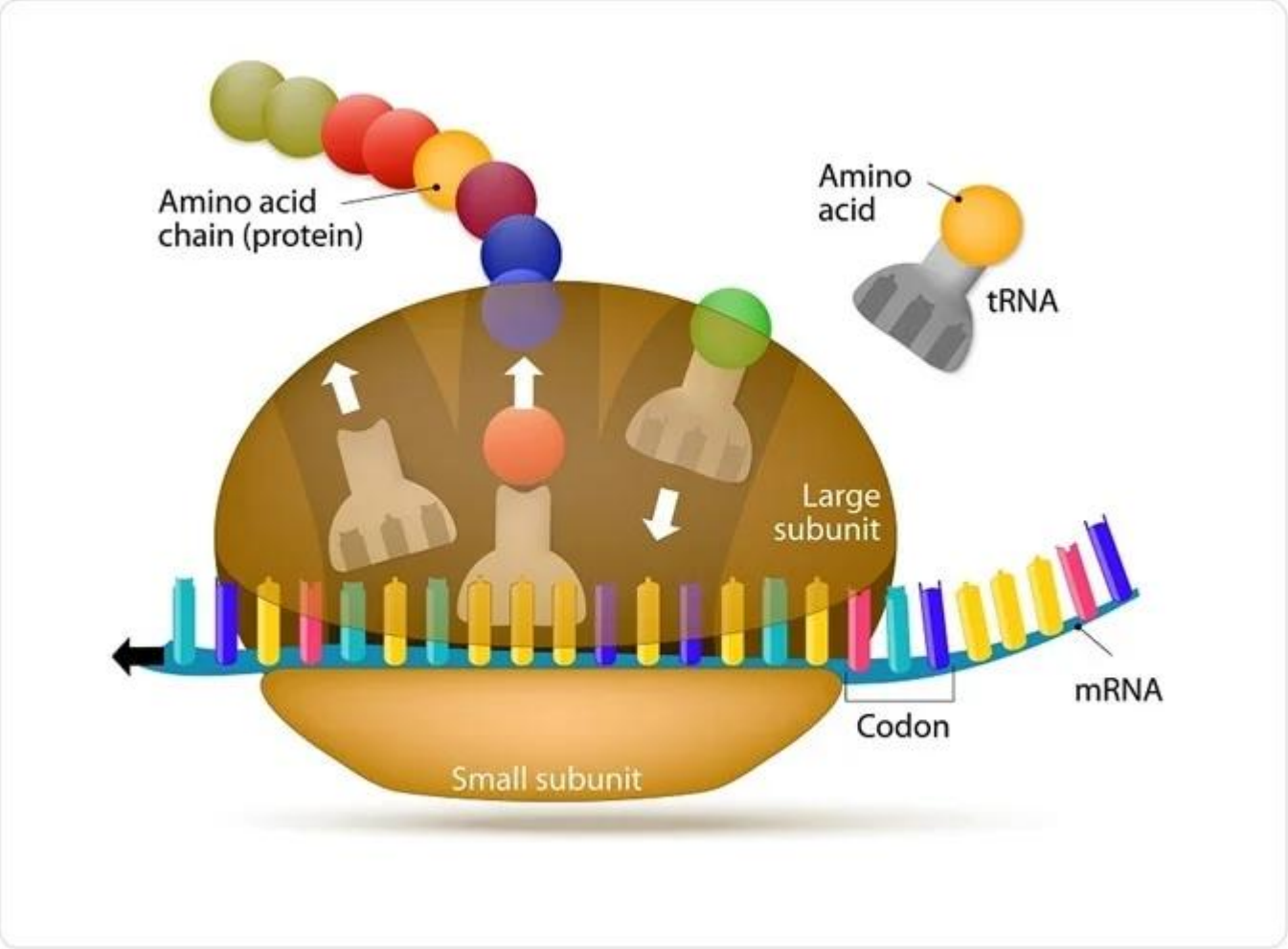
西交利物浦大學

Protein Folding

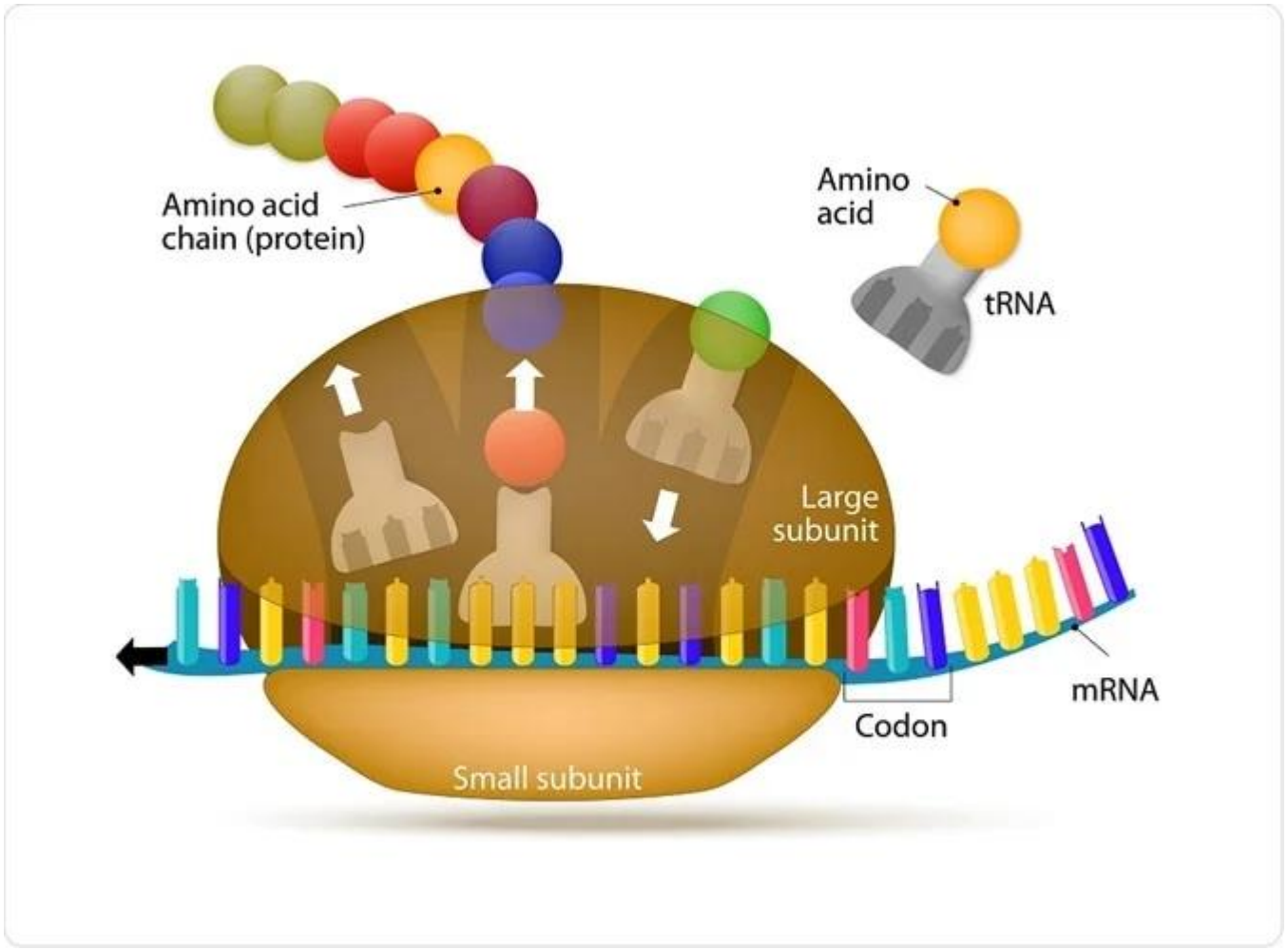


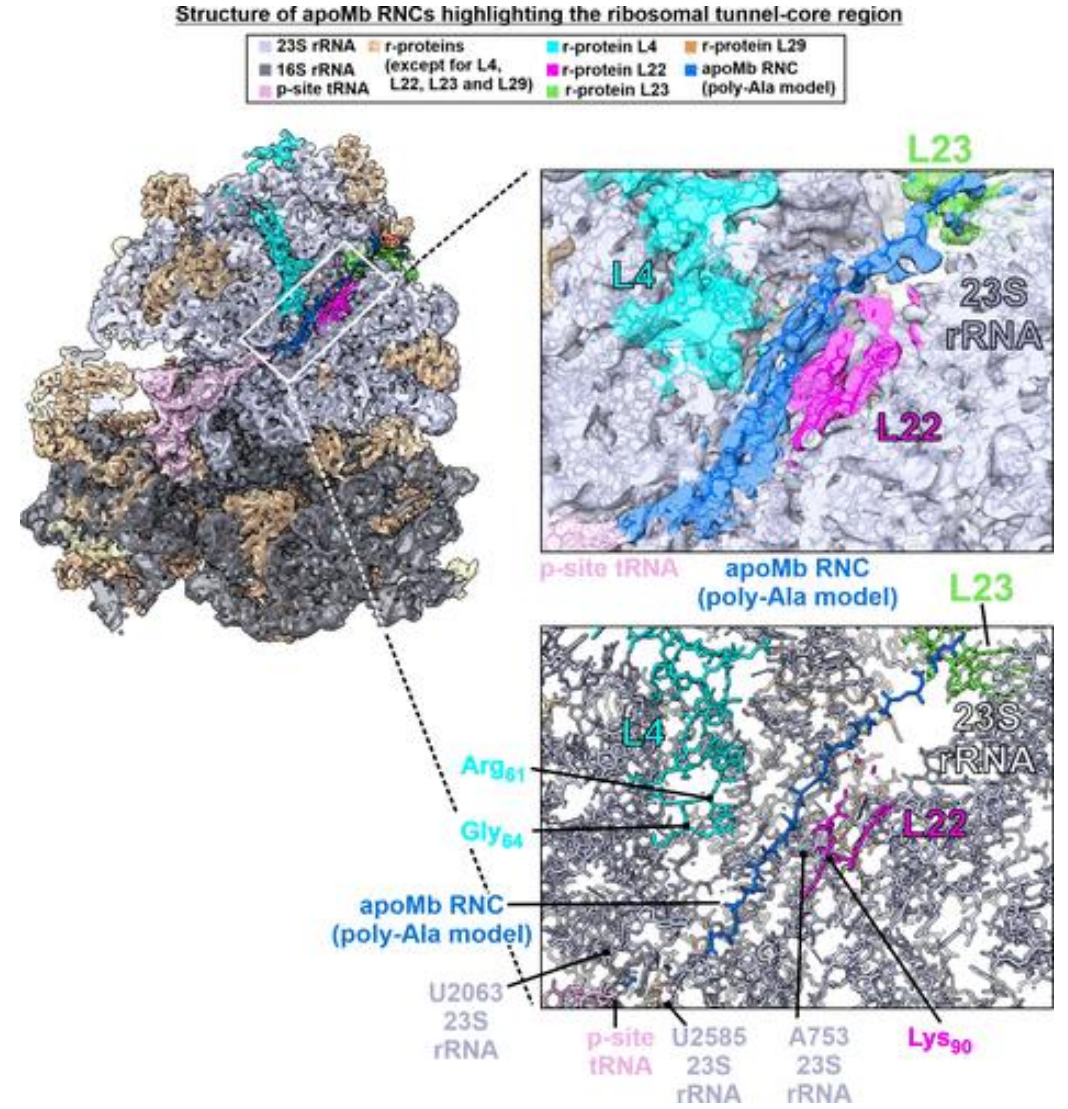
- ✓ The terms folding can be seen as a **noun** (the "folding" in the sense of the "folded state" of the protein) or as a **verb** (the "folding process").
- ✓ The folded state is sufficient to understand how the protein should work, and how mutations affects the protein.
- ✓ The process can be used to understand misfolding and the Physical principles which brings to folding.

The ribosome and protein synthesis



The ribosome and protein synthesis



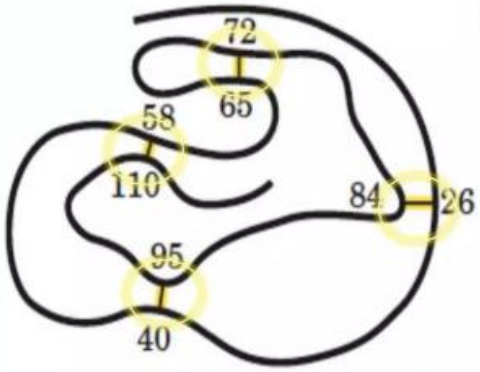


ARTICLE | February 1, 2024

Mapping Protein–Protein Interactions at Birth: Single-Particle Cryo-EM Analysis of a Ribosome–Nascent Globin Complex

Meranda M. Masse, Rachel B. Hutchinson, Christopher E. Morgan, Heather J. Allaman, Hongqing Guan, Edward W. Yu, and Silvia Cavagnero*

Anfisen's experiment

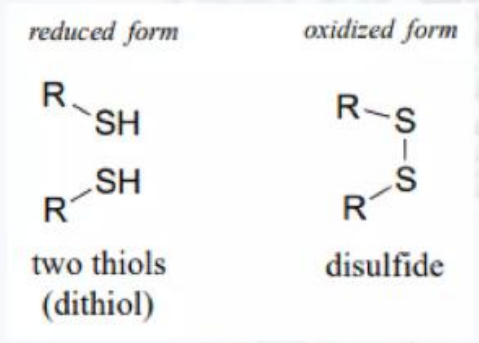


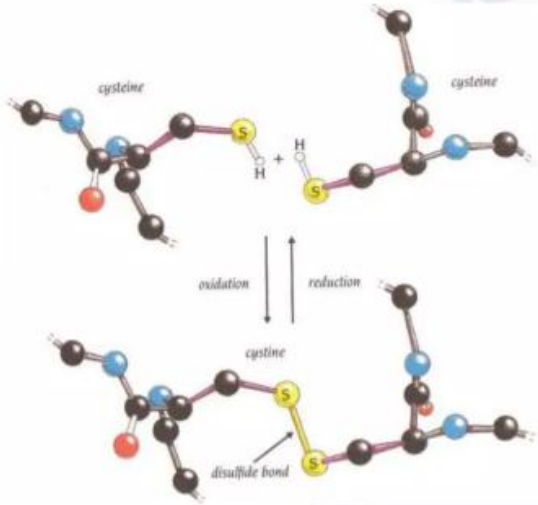
Native state of ribonuclease A

Native state; catalytically active.

Sulfhydryl group pairings in native conformation

58-110
72-65
26-84
40-95





Two cysteines in proximity can form a covalent bond called disulfide linkage.

SH groups are oxidized to form disulfide bonds.

2-MERCAPTOETHANOL

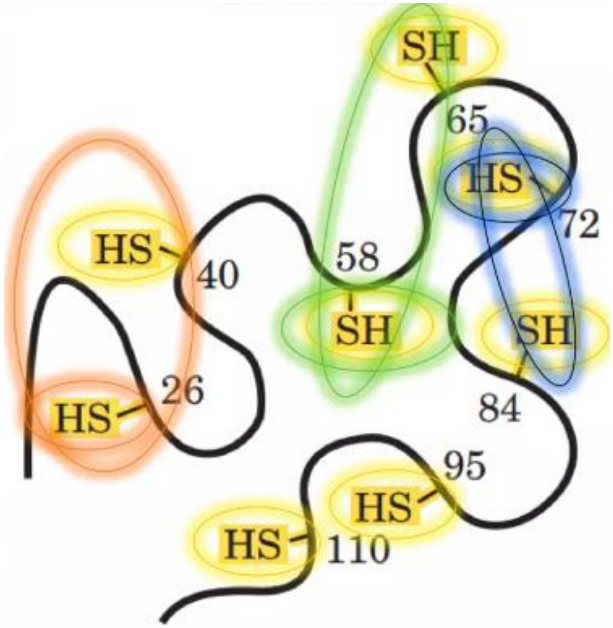
2-mercaptoethanol cleaves the disulfide bonds that may form between thiol groups of cysteine residues.

Urea

Urea interacts with the polar regions of the proteins resulting in the unfolding of the tertiary and secondary structure of the proteins and exposure of the hydrophobic core to water and urea, simply put- urea denatures proteins



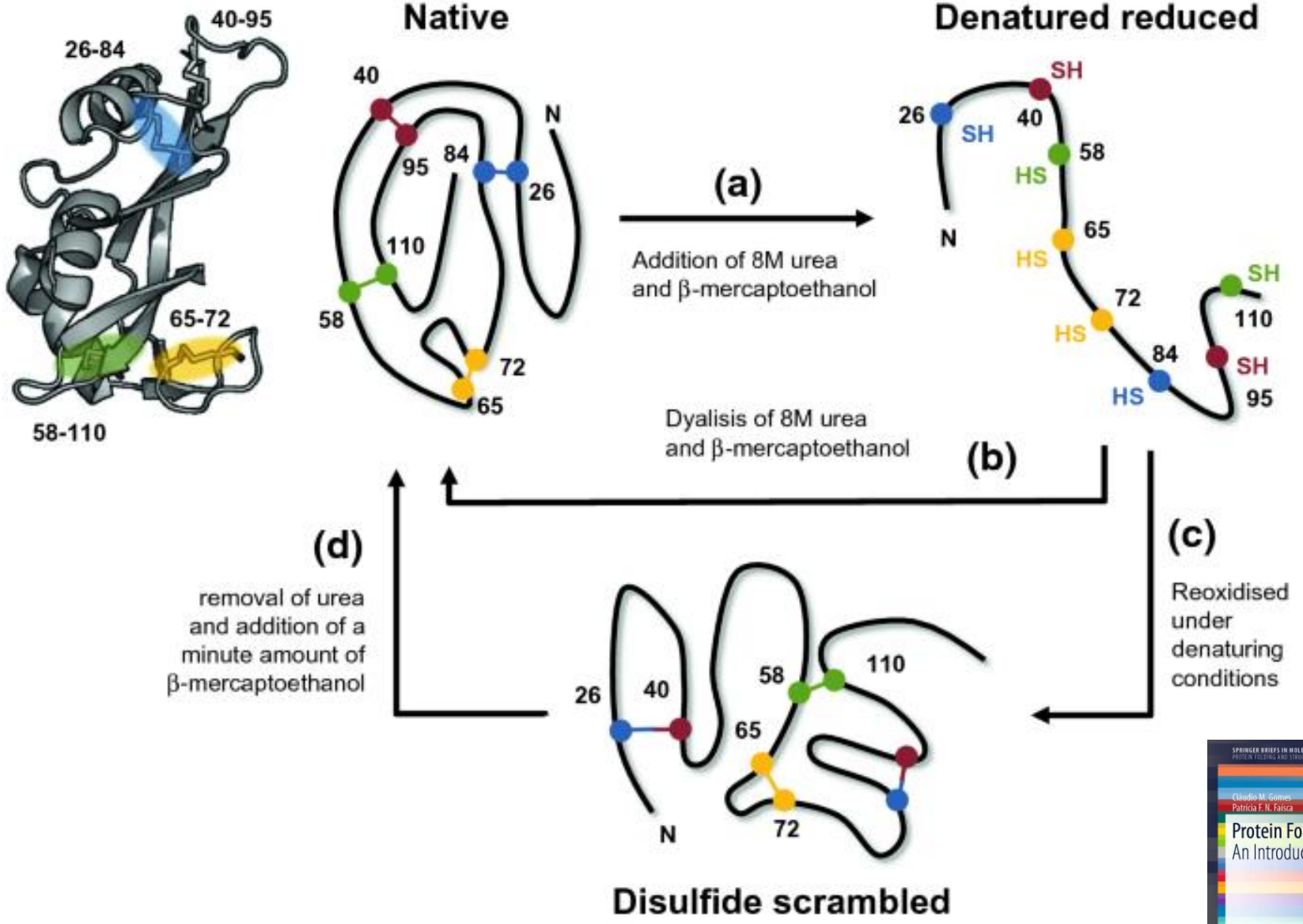
Anfisen's experiment

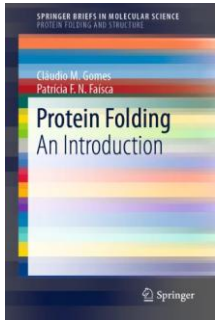


Lehninger Principles of Biochemistry, 8th Edition

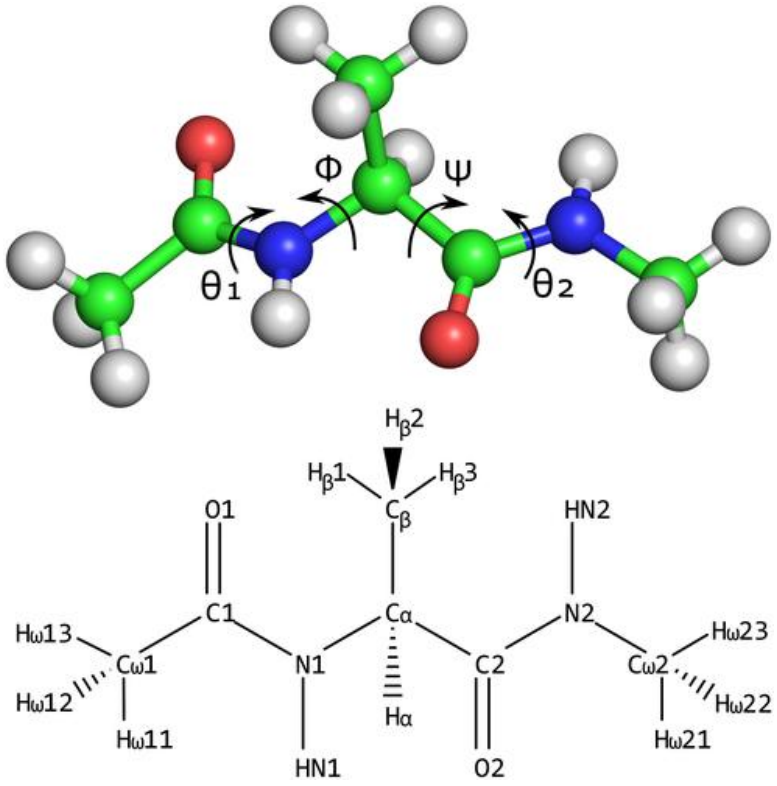
$7 \times 5 \times 3 \times 1 = 105$

Number of theoretical configuration in which ribonuclease can refold





Levinthal's paradox



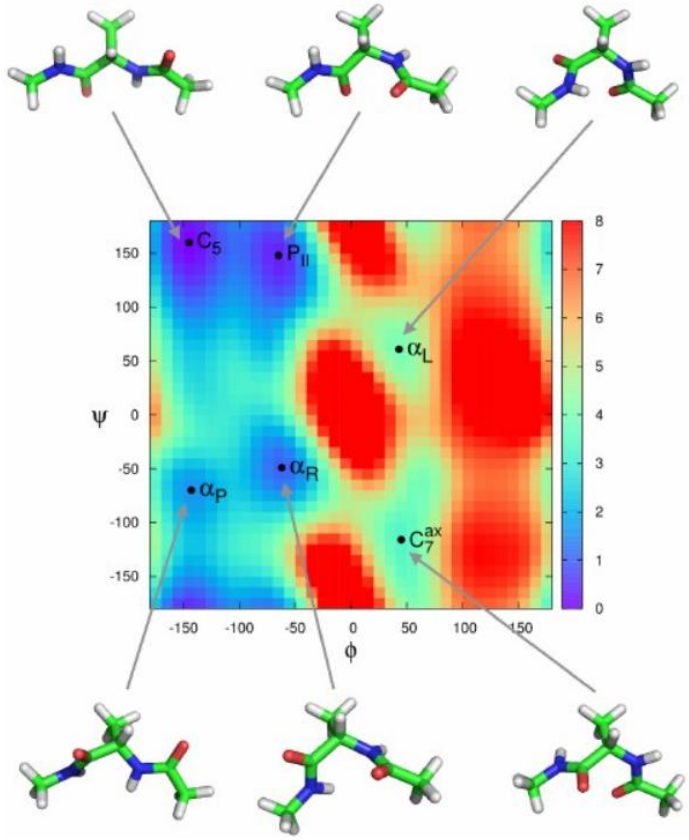


Figure 1: Ramachandran map of the alanine dipeptide in its (φ, ψ) space, with the locations of six local energy minima and their corresponding conformations (Chodera et al. 2006; Jaillet et al. 2011).

Protein length: 100 a.a.

Number of Ramachandran angles: 200

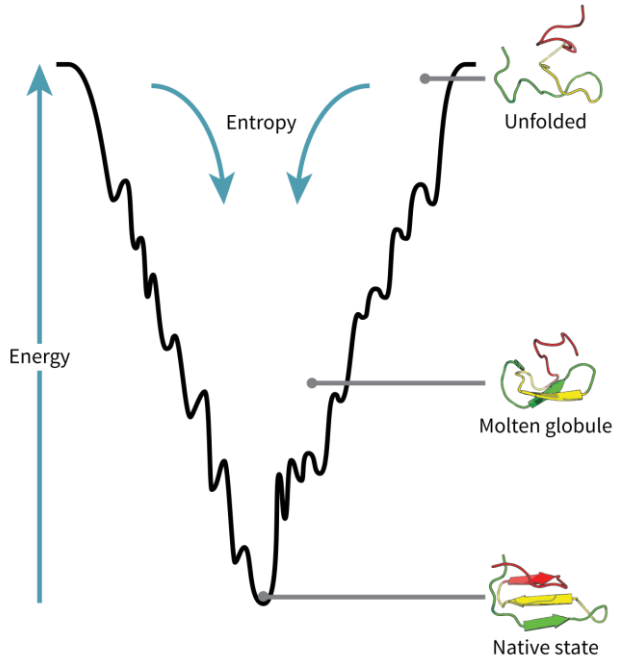
Possible state per each angle (average): 3

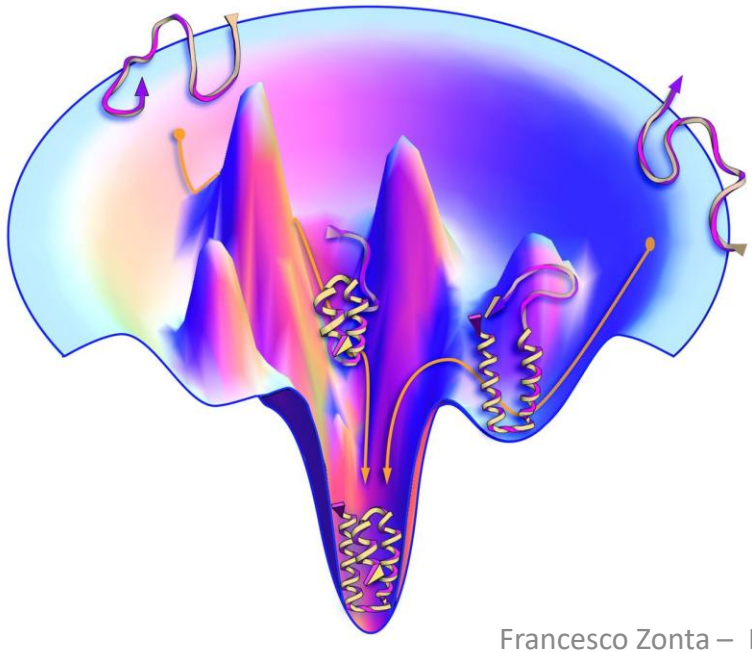
Number of possible combinations: $3^{200} = 2.7 \cdot 10^{95}$

Age of the universe (in seconds) : $4.35 \cdot 10^{17}$

Levinthal's paradox: Protein cannot explore the whole configuration space to find the native state (minimum of energy).

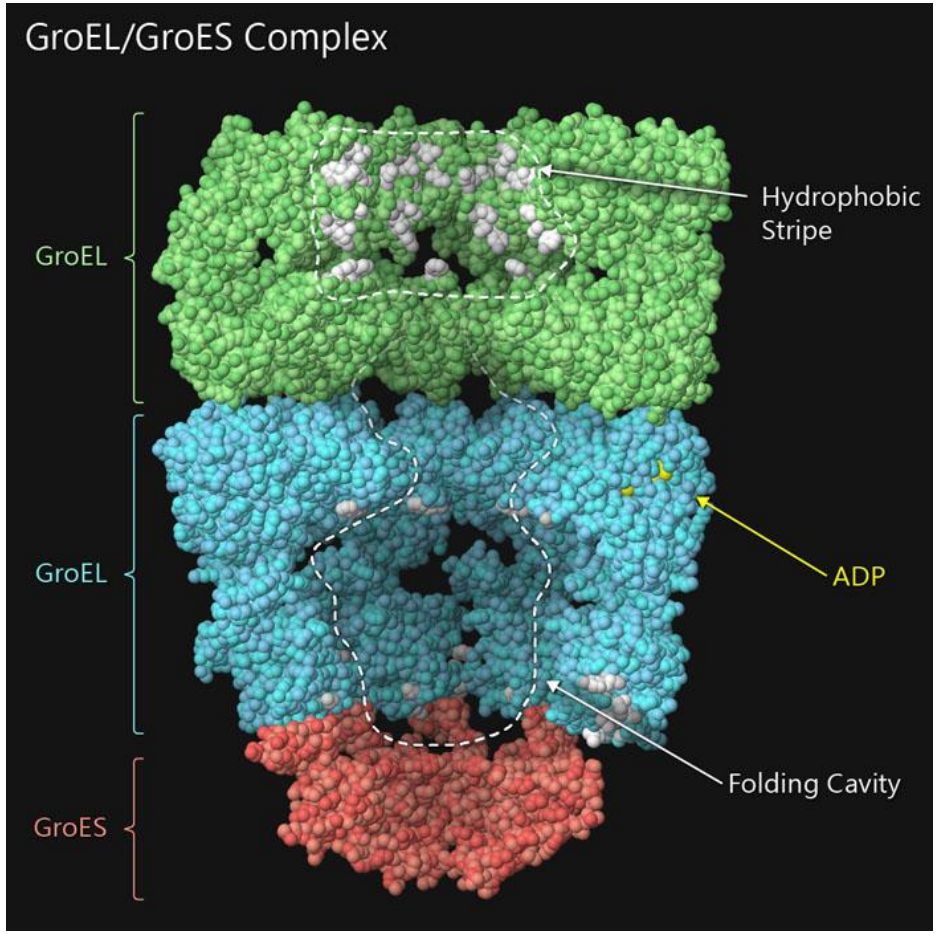
The protein folding funnel





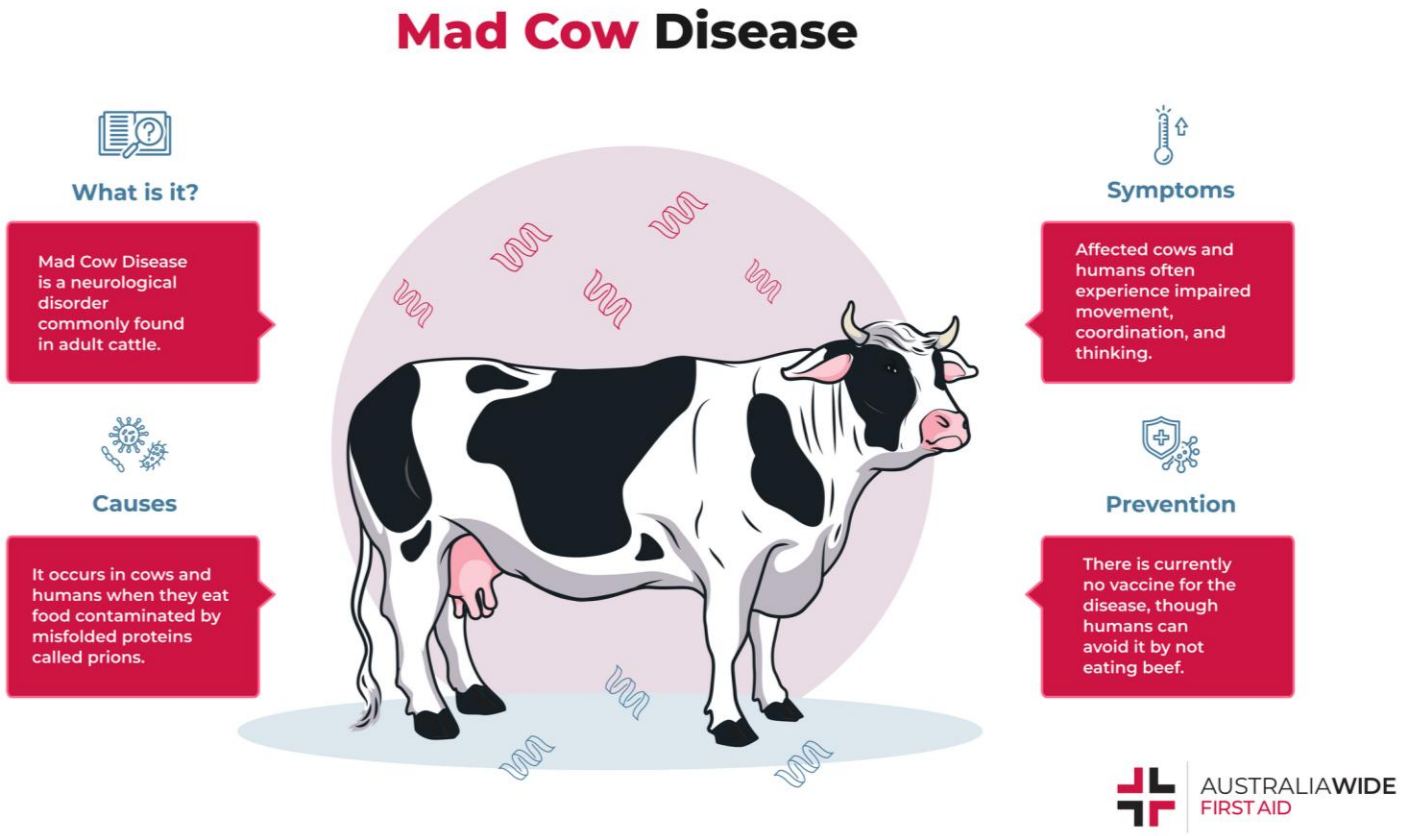
- A possible way to solve Levinthal's paradox is assuming that the energy landscape is shaped as a funnel.
- The funnel shape represents a bias toward the native state, with energy decreasing as the protein approaches its correctly folded structure.
- Multiple folding pathways can lead to the same native state, eliminating the need to search through all possible conformations randomly.
- The width of the funnel represents conformational entropy (number of possible states), which decreases as folding progresses and the structure becomes more ordered.
- Local minima in the funnel can act as folding intermediates or misfolded traps, but the overall downward slope drives folding toward the global energy minimum.
- Folding is primarily driven by the hydrophobic effect, where hydrophobic residues collapse into the protein core to minimize unfavorable contacts with water, while hydrophilic residues remain on the surface.

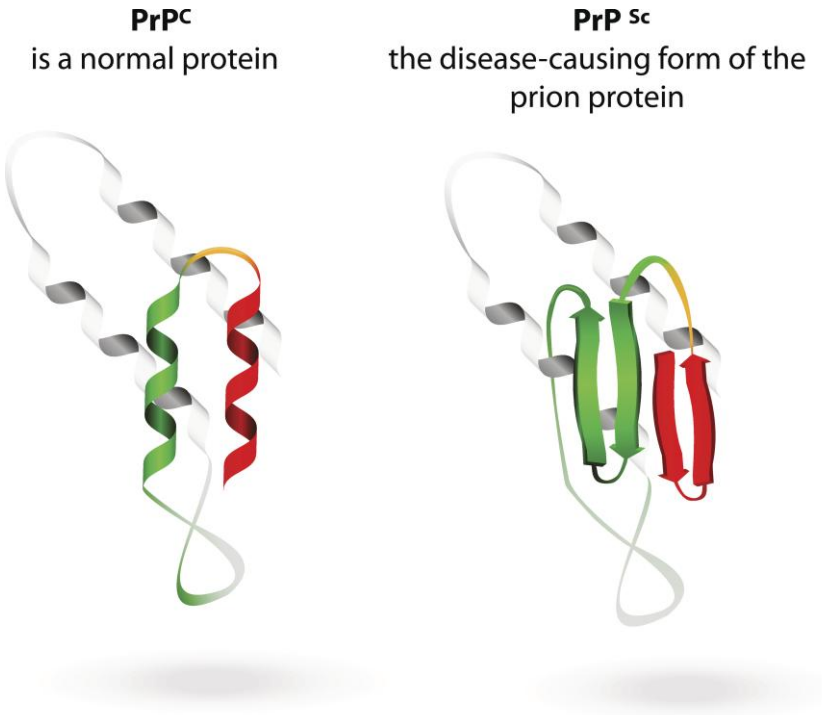
Chaperons

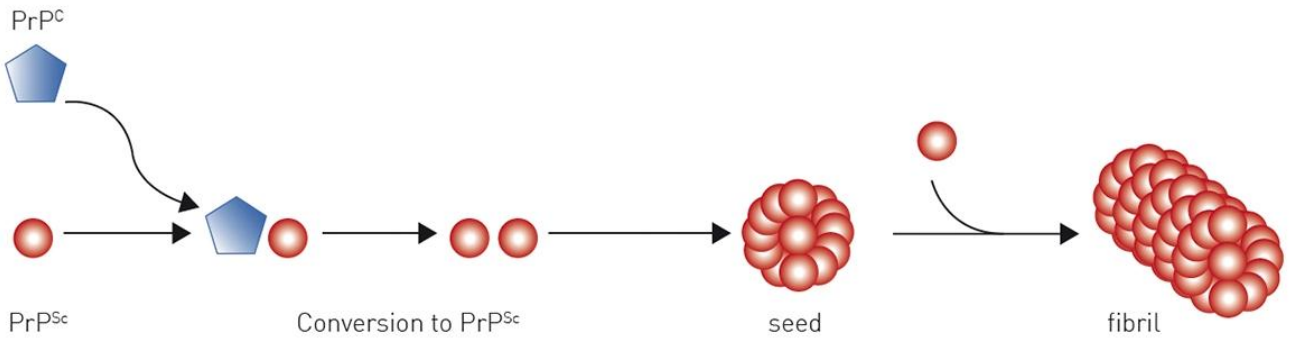


- Molecular chaperones are proteins that assist in the proper folding of other proteins without being part of the final structure.
- Chaperones prevent aggregation by binding to exposed hydrophobic regions on unfolded or misfolded proteins, shielding them from inappropriate interactions.
- Chaperonins are a specific class of chaperones that form barrel-shaped chambers where proteins can fold in isolation from the crowded cellular environment.
- ATP hydrolysis provides energy for chaperone cycles of binding and releasing substrate proteins, giving them multiple chances to reach their native state.
- Examples include Hsp70 (general chaperone) and GroEL/GroES (chaperonin system in bacteria), which help proteins navigate their folding landscape efficiently.

Misfolding and aggregations







Prions are misfolded proteins that can act as infectious agents by inducing normal proteins of the same type to adopt their aberrant conformation, leading to self-propagating aggregation and neurodegenerative diseases like Creutzfeldt-Jakob disease.

A more accurate version of the protein folding funnel

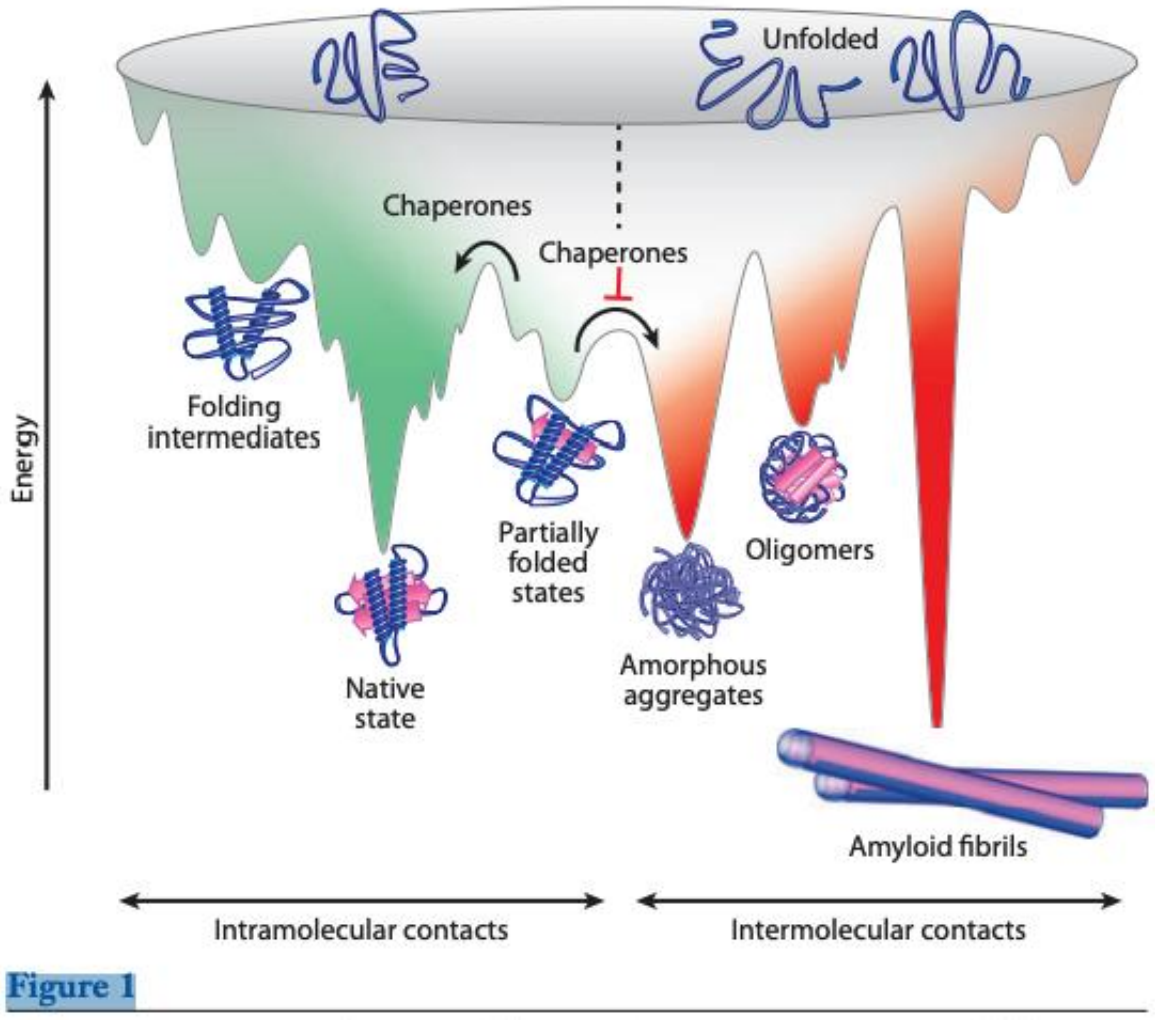
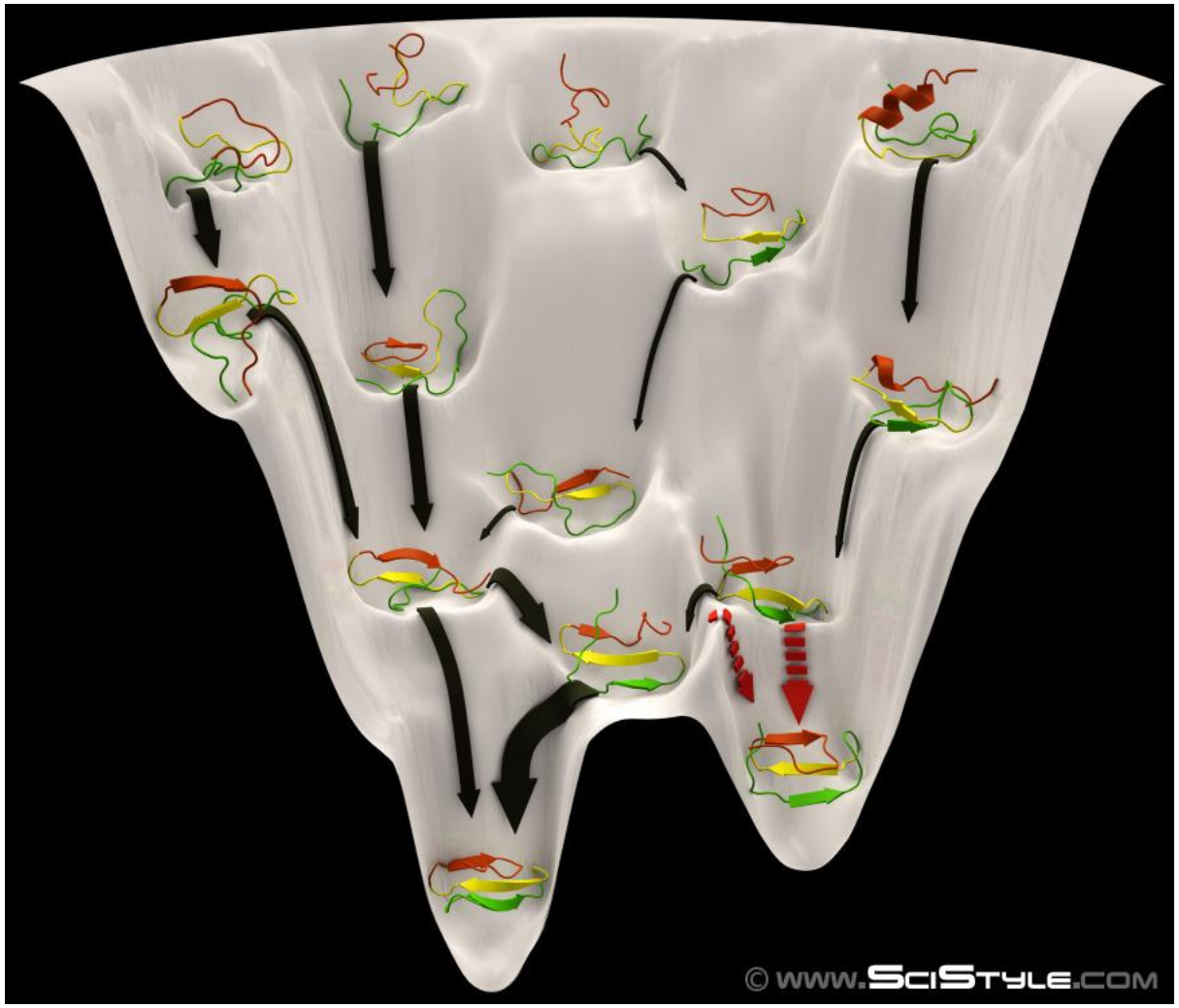


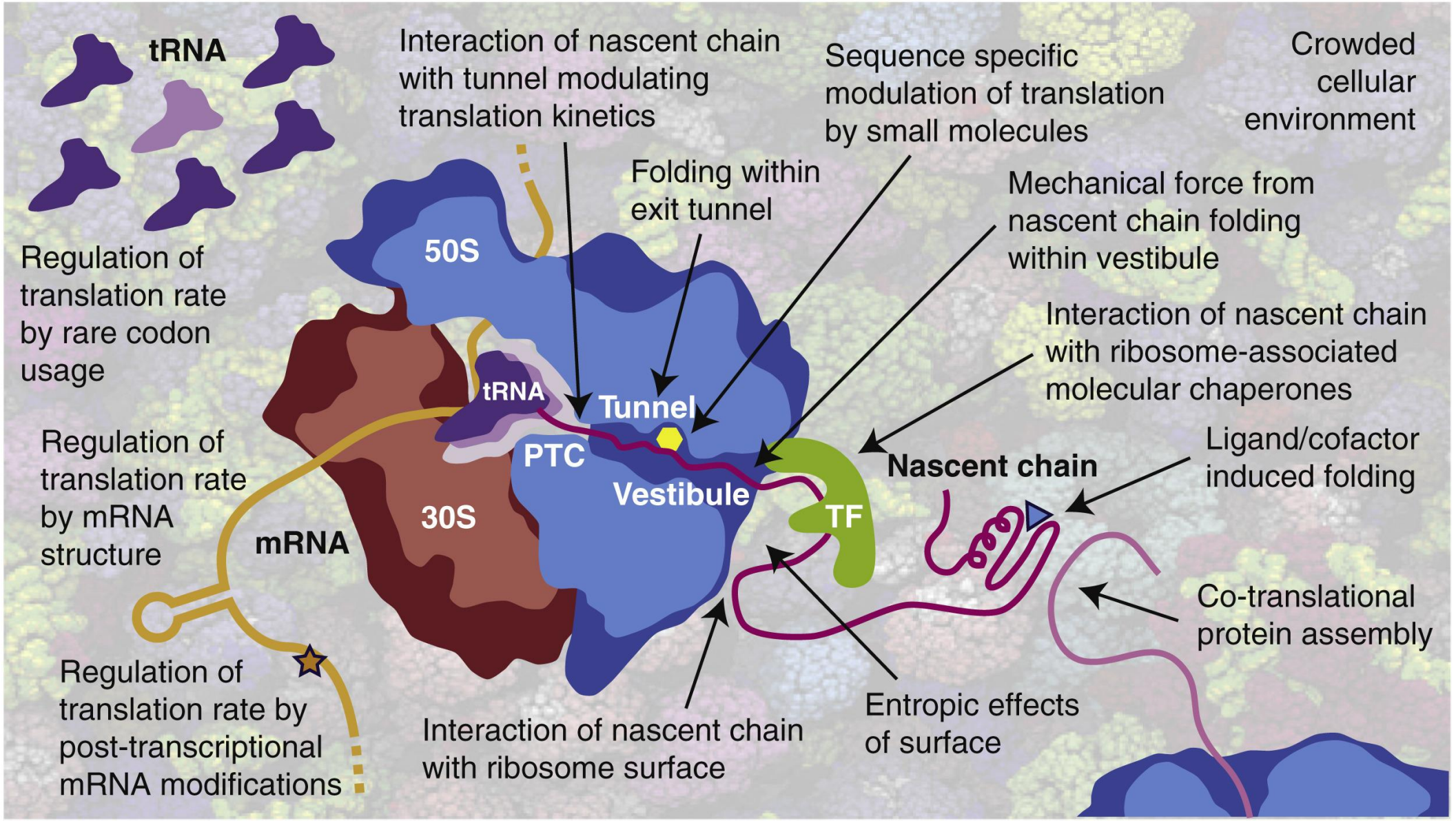
Figure 1



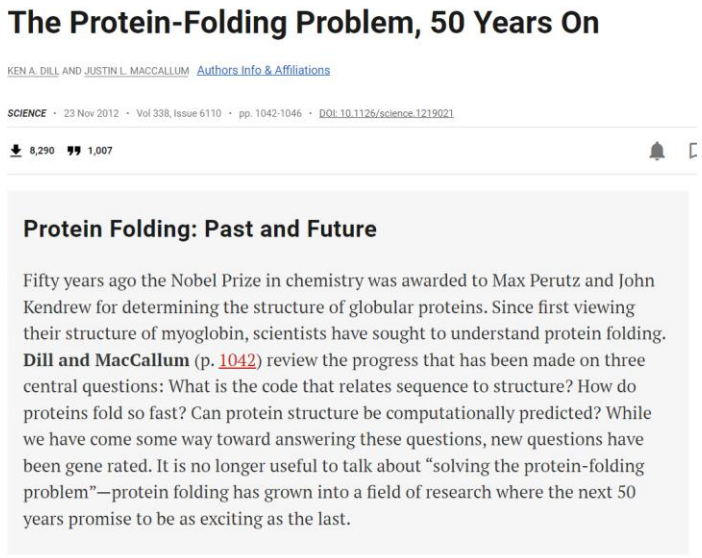
REVIEW · Volume 44, Issue 11, P914-926, November 2019 · *Open Access* Download Full Issue

Nature and Regulation of Protein Folding on the Ribosome

Christopher A. Waudby [1] · Christopher M. Dobson [2] · John Christodoulou [1]



Protein Folding and structural predictions



The Protein-Folding Problem, 50 Years On

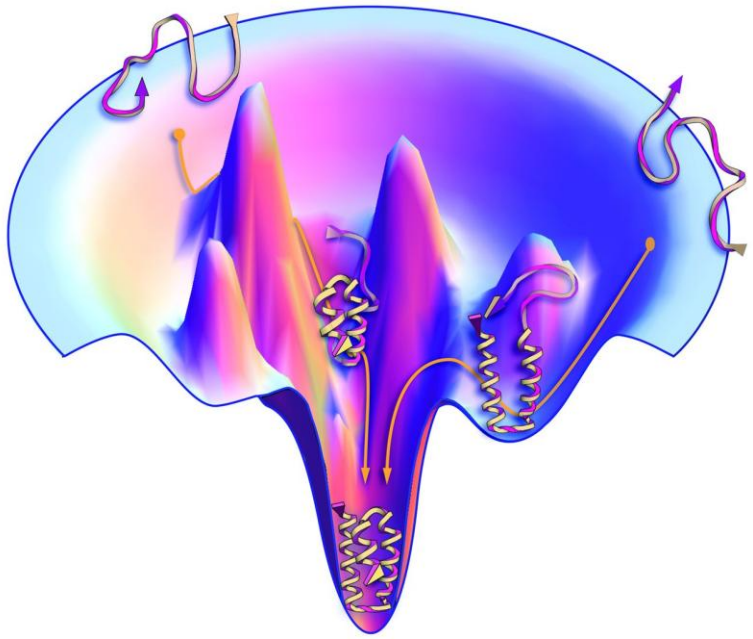
KEN A. DILL AND JUSTIN L. MACCALLUM Authors Info & Affiliations

SCIENCE · 23 Nov 2012 · Vol 338, Issue 6110 · pp. 1042-1046 · DOI: 10.1126/science.1219021

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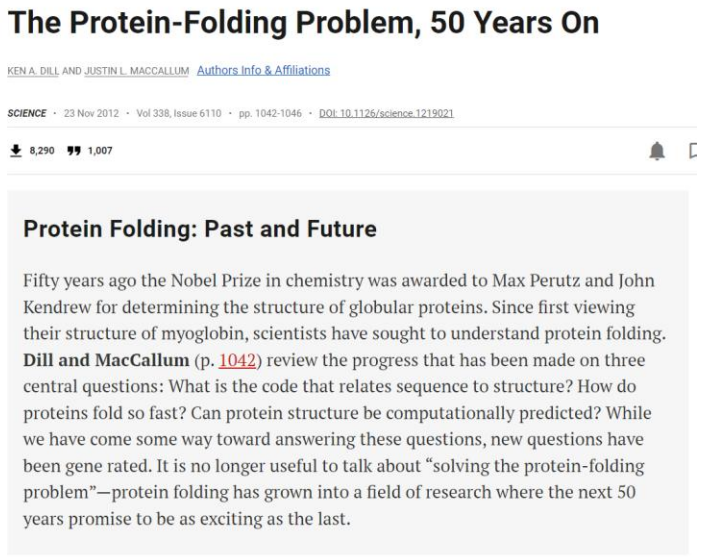
Protein Folding: Past and Future

Fifty years ago the Nobel Prize in chemistry was awarded to Max Perutz and John Kendrew for determining the structure of globular proteins. Since first viewing their structure of myoglobin, scientists have sought to understand protein folding. **Dill and MacCallum** (p. 1042) review the progress that has been made on three central questions: What is the code that relates sequence to structure? How do proteins fold so fast? Can protein structure be computationally predicted? While we have come some way toward answering these questions, new questions have been gene rated. It is no longer useful to talk about "solving the protein-folding problem"—protein folding has grown into a field of research where the next 50 years promise to be as exciting as the last.



- ✓ Predicting the three-dimensional structure of a protein from its amino acid sequence is a fundamental challenge in structural biology.
- ✓ This problem, known as the "protein folding problem," has been a long-standing and complex research area.
- ✓ Understanding protein folding is crucial for elucidating protein function, disease mechanisms, and drug design.
- ✓ The protein folding problem can be approached from two perspectives: a. Predicting the final folded structure (the native state) b. Predicting the folding process and pathway
- ✓ The folding process involves a complex series of conformational changes and intermediate states as the protein transitions from an unfolded state to its native structure.

Methods for Structural predictions



The Protein-Folding Problem, 50 Years On

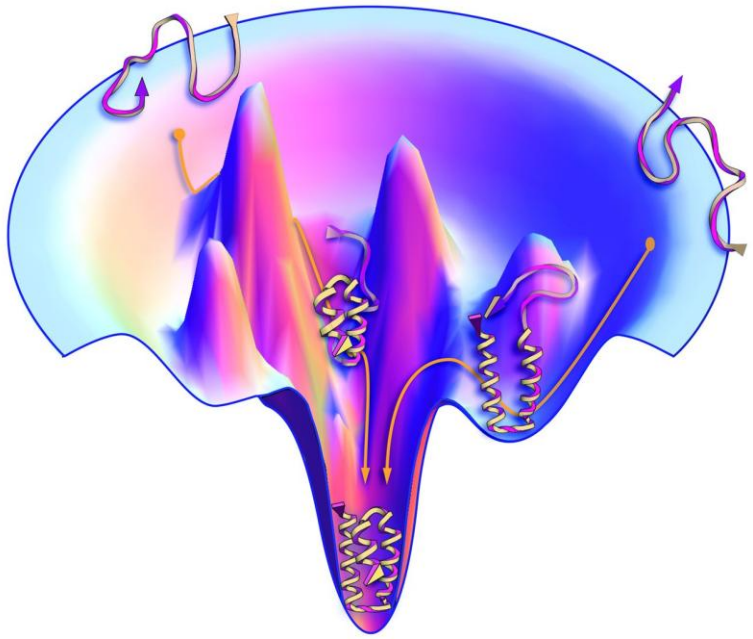
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SCIENCE · 23 Nov 2012 · Vol 338, Issue 6110 · pp. 1042-1046 · DOI: 10.1126/science.1219021

8,290 1,007

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Homology Modeling:

Homology modeling is a technique for predicting the three-dimensional structure of a protein based on its sequence similarity to other proteins with known structures.

It relies on the principle that proteins with similar sequences often have similar structures.

Software tools like MODELLER, SWISS-MODEL, and I-TASSER are widely used for homology modeling.

Threading or Fold Recognition:

Threading methods attempt to recognize known protein folds or structural motifs in the target protein sequence.

They involve aligning the target sequence onto known protein structures and evaluating the compatibility based on various scoring functions.

Examples of threading programs include PROSPECTOR, SPARK, and pGenTHREADER.

Ab initio or de novo Modeling:

Ab initio methods aim to predict protein structures from their amino acid sequences without relying on known structural templates.

These methods often use physics-based energy functions or machine learning techniques to sample the conformational space and identify low-energy structures.

Popular ab initio modeling programs include ROSETTA, QUARK, and UNRES.

Machine Learning and Deep Learning Methods:

Machine learning and deep learning techniques, such as neural networks and deep belief networks, have recently gained popularity in protein structure prediction.

These methods can learn patterns and features from large datasets of known protein structures and sequences.

Examples include AlphaFold, trRosetta, and the methods used in the CASP competitions.

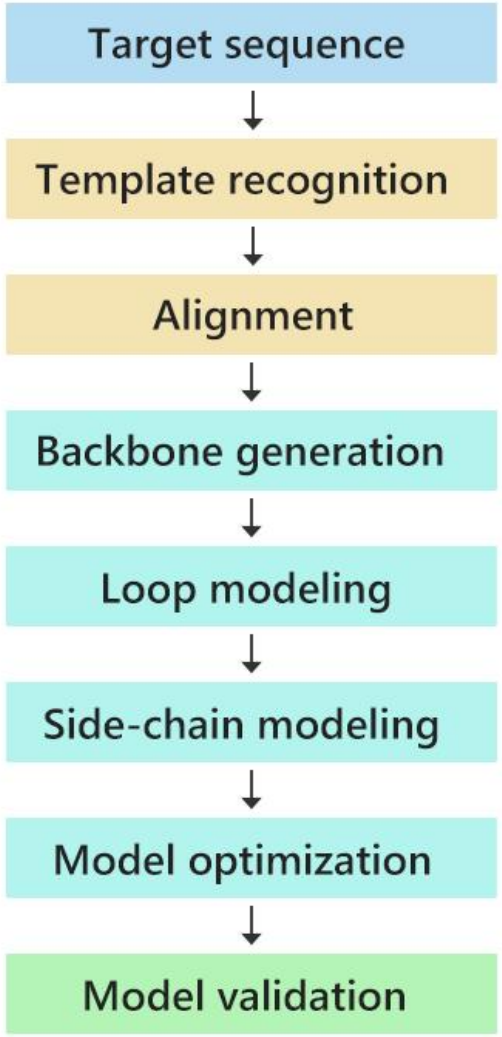
Hybrid Methods:

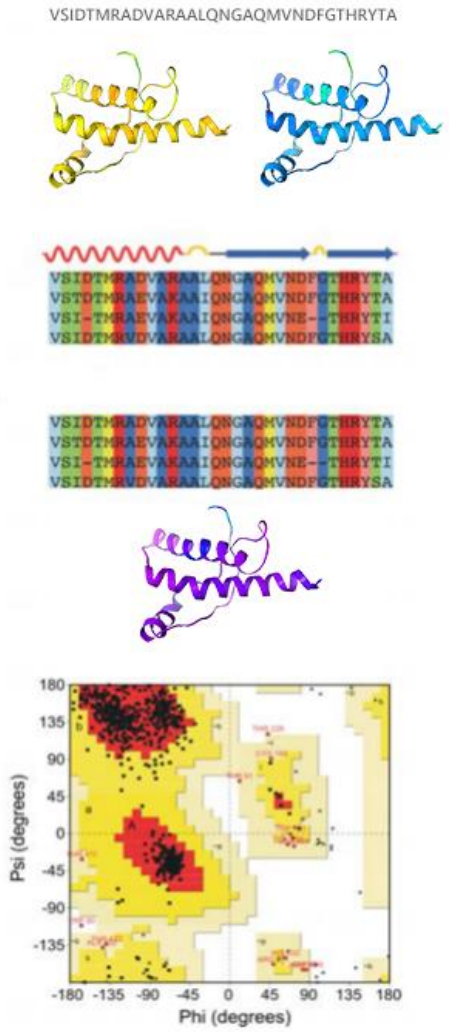
Hybrid methods combine various computational techniques, such as homology modeling, threading, ab initio modeling, and machine learning, to improve the accuracy of structure prediction.

These methods often employ consensus or meta-server approaches, integrating the results from multiple structure prediction methods.

Examples include MULTICOM, I-TASSER, and Robetta.

Homology Modeling

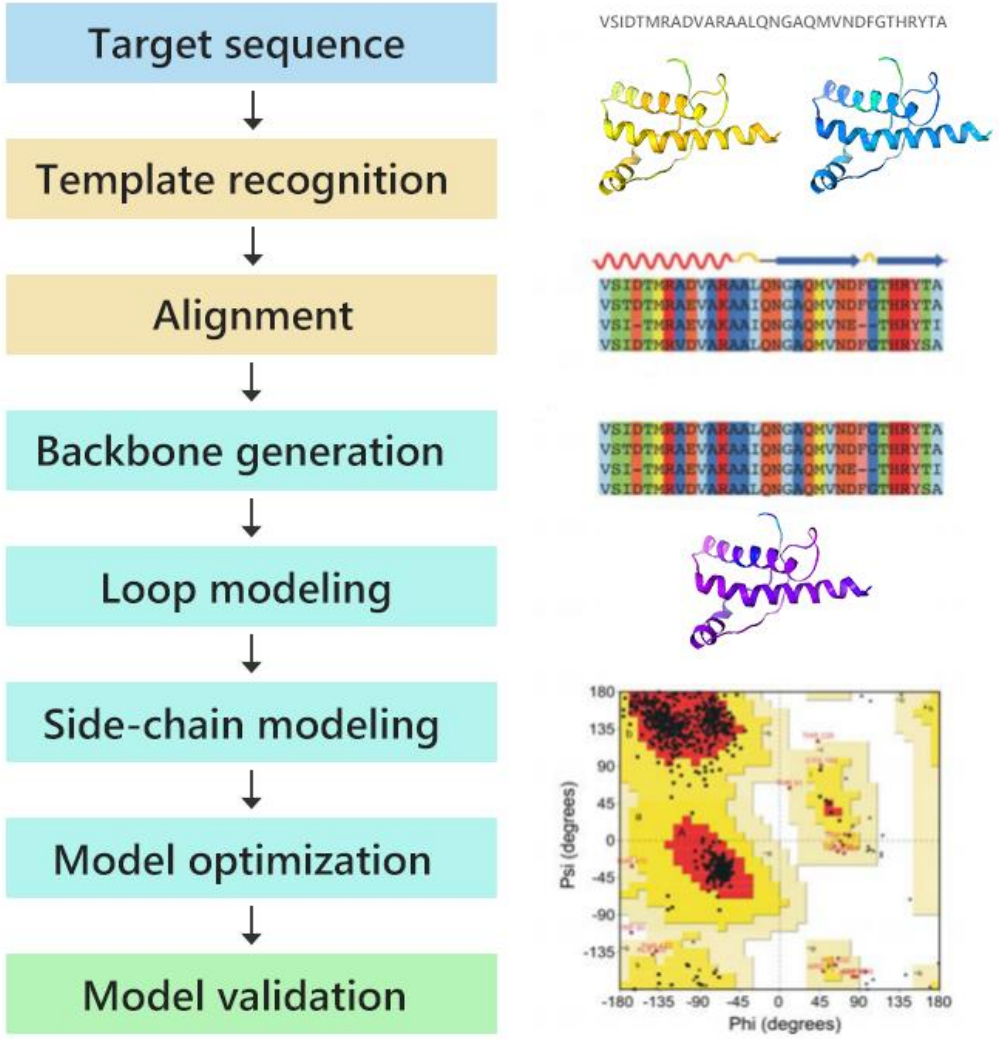




Homology modeling, also known as comparative modeling, uses homologous sequences with known 3D structures for the modeling and prediction of the structure of a target sequence

- Basic assumption: two homologous proteins will share very similar structures
- Although the number of actual proteins is vast, there is a limited set of tertiary structural motifs to which most proteins belong
- Homology modeling is one of the best-performing methods and (in most cases) gives accurate predicted models

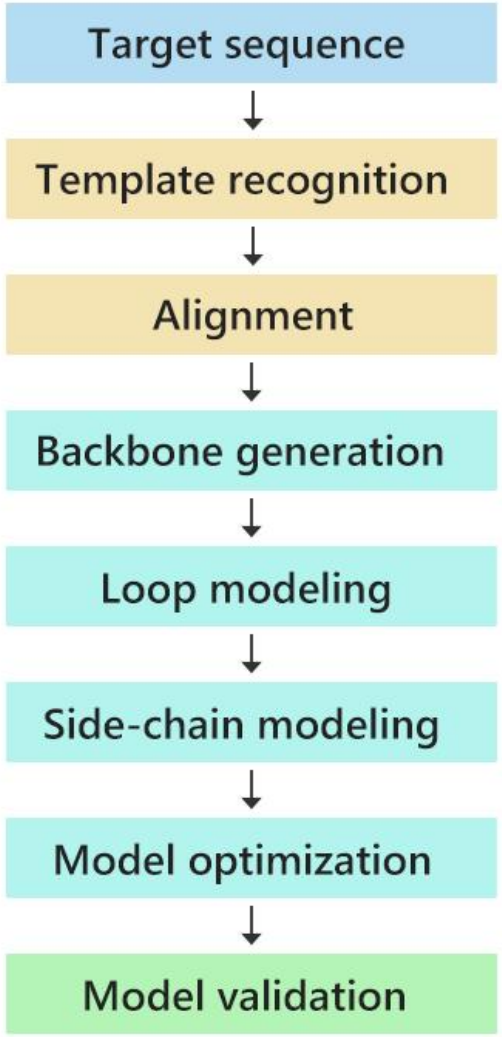
Homology Modeling

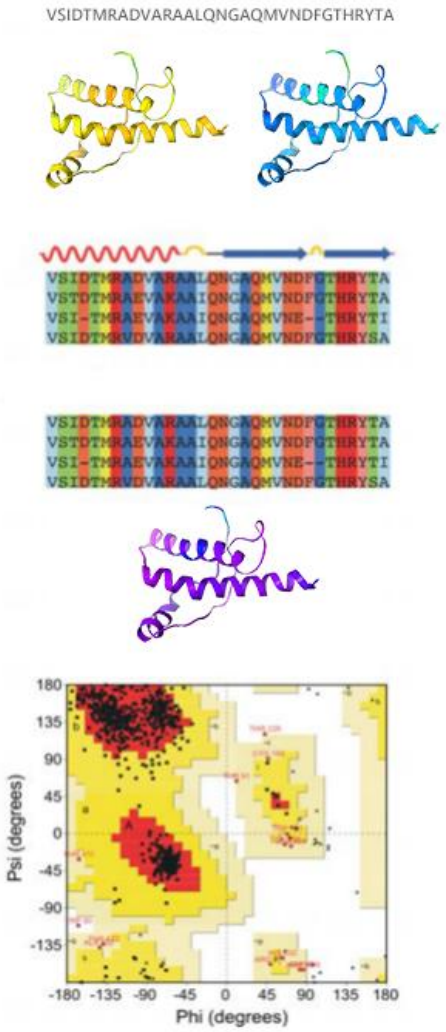


Homology modeling relies on

- The identification of one or more known protein structures likely to resemble the structure of the query sequence
- The 3D structure of the template must be determined by reliable empirical methods such as crystallography or NMR, and is typically a published atomic coordinate file from PDB.
- The production of an alignment that maps residues in the query sequence to residues in the template sequence

Homology Modeling



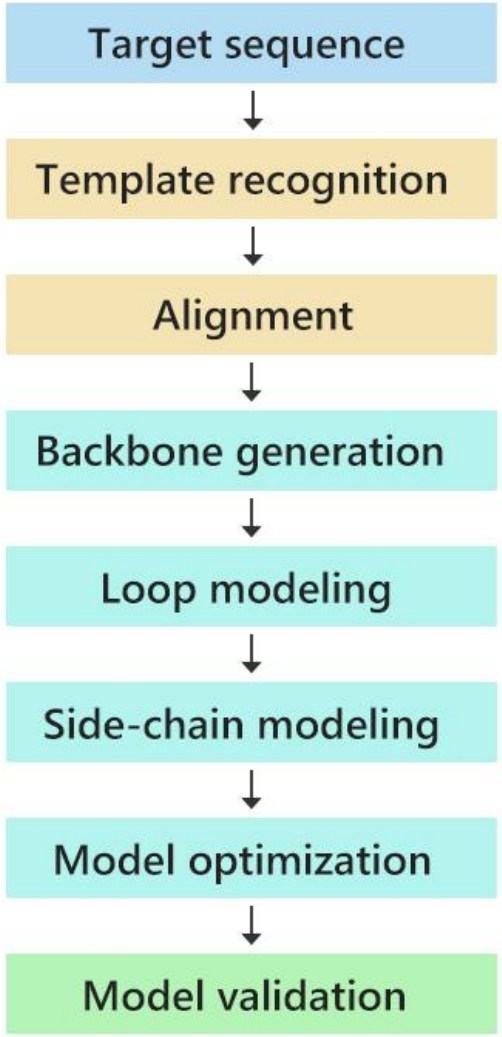


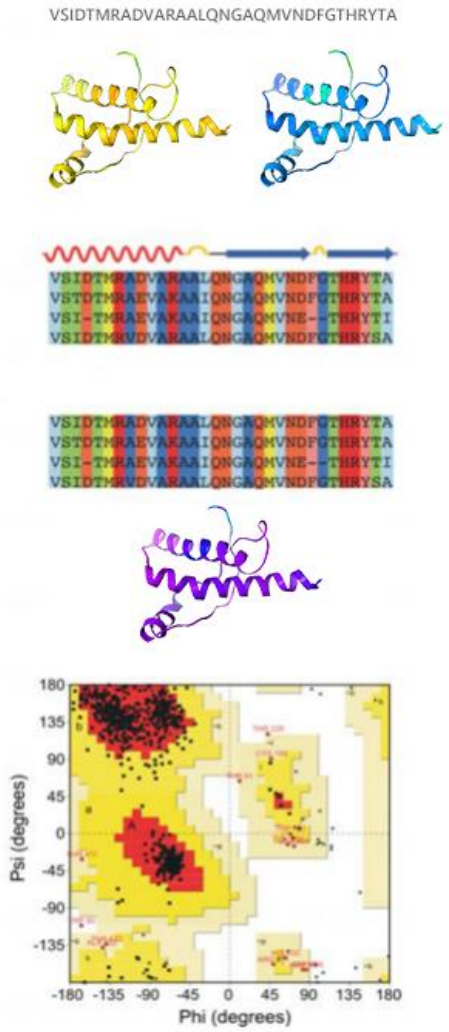
Step required to perform Homology Modeling:

1. Retrieving the sequence to be modeled (target sequence)
2. Identifying suitable templates from PDB using BLAST
3. Aligning target sequence with template sequences
4. Performing homology modeling using proper tools
5. Model quality check
6. Quantitative comparison of the model with the experimental information

(In most cases, all these steps are done automatically by the software)

Homology Modeling





2. Identifying suitable templates from PDB using BLAST

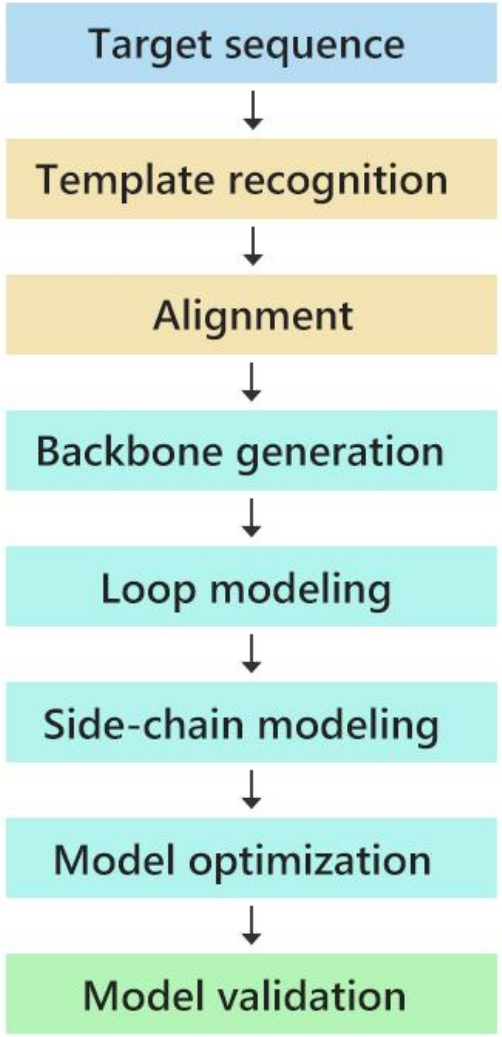
Go for the best hit (>30 - 40% identity)

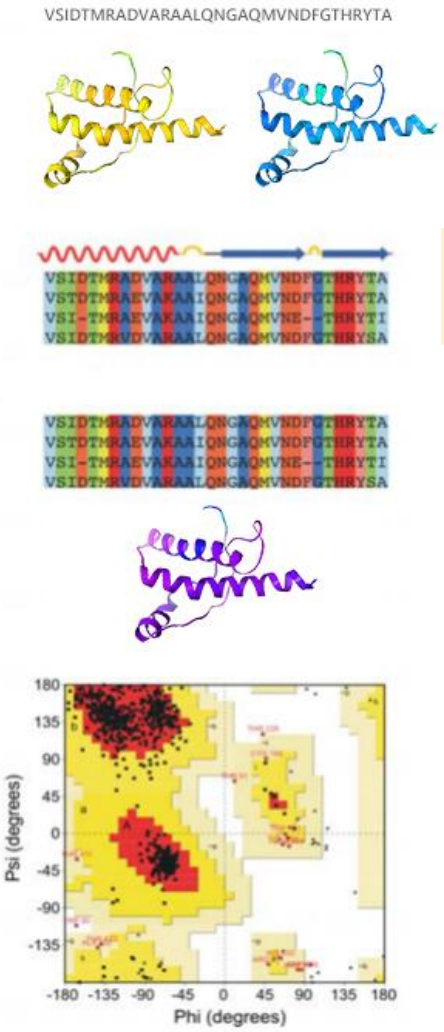
30% identity → 70% Accuracy

50% identity → 90% Accuracy

70% identity → Almost perfect model

Homology Modeling



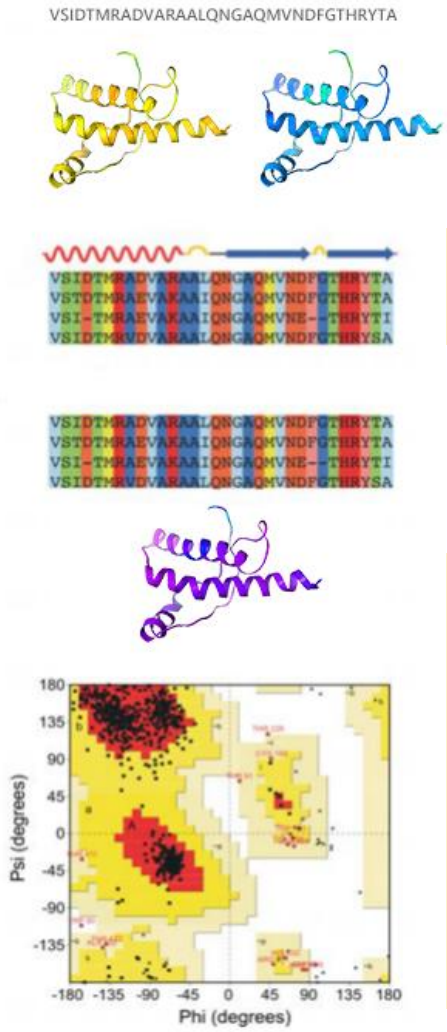


3. Aligning target sequence with template sequences

Remove gapped positions
Remove residues in target that are missing in template

Homology Modeling

- Target sequence
- Template recognition
- Alignment
- Backbone generation
- Loop modeling
- Side-chain modeling
- Model optimization
- Model validation



3. Aligning target sequence with template sequences

Remove gapped positions
Remove residues in target that are missing in template

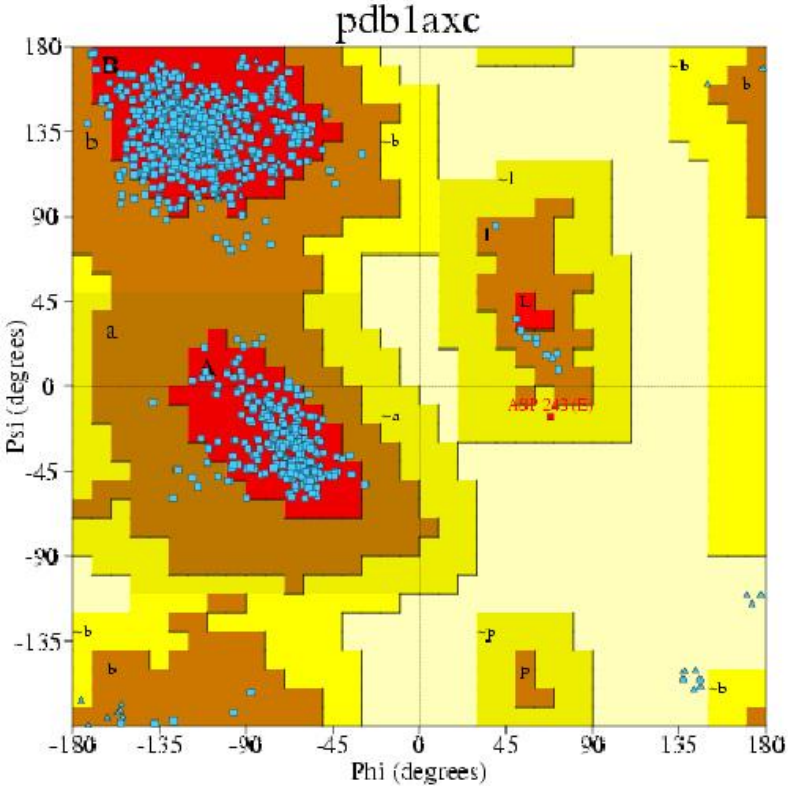
4. Performing homology modeling using proper tools

First, the homology modeling routine arranges the backbone identically to that of the template.

This means not only the positions of alpha carbons, but also the phi and psi angles and secondary structure are made identical to the template.

Next, side chain positions are adjusted to minimize collisions.

Homology Modeling



5. Model quality check

More than 90% of residues are in the most favorable regions of the Ramachandran plot

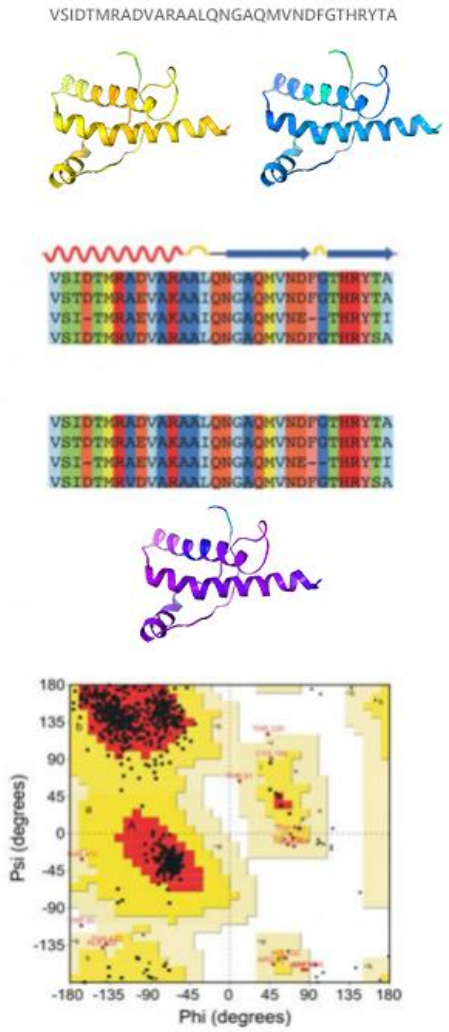
There are very few unusual side chain angles, as well as main chain bond lengths and angles with G-factors < -0.5 (unusual) and G-factors < -1 (highly unusual)

There are low deviations from the ideal covalent geometry (based on the difference values in the Distorted geometry plot)

https://www.ebi.ac.uk/thornton-srv/software/PROCHECK/

Homology Modeling

Target sequence
↓
Template recognition
↓
Alignment
↓
Backbone generation
↓
Loop modeling
↓
Side-chain modeling
↓
Model optimization
↓
Model validation



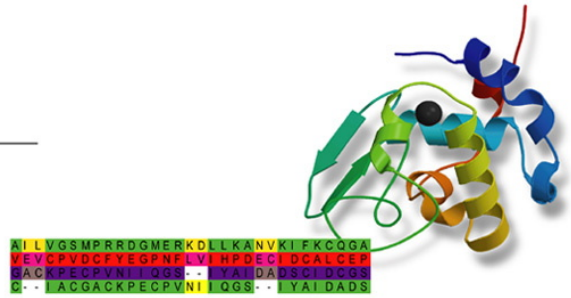
6. Quantitative comparison of the model with the experimental information

0.0-0.5 Å	→	Essentially Identical
<1.5 Å	→	Very good fit
< 5.0 Å	→	Moderately good fit
5.0-7.0 Å	→	Structurally related
> 7.0 Å	→	Dubious relationship
> 12.0 Å	→	Completely unrelated

Modeller

Modeller

Program for Comparative Protein Structure Modelling by Satisfaction of Spatial Restraints





https://salilab.org/modeller/

Satisfaction of spatial restraints - MODELLER uses spatial restraints derived from the template structure and alignment, optimizing the model to satisfy distance and angle constraints rather than just copying coordinates, leading to better stereochemistry.

Handles multiple templates - Can simultaneously use information from multiple template structures, combining the best features from different homologs rather than being limited to a single template.

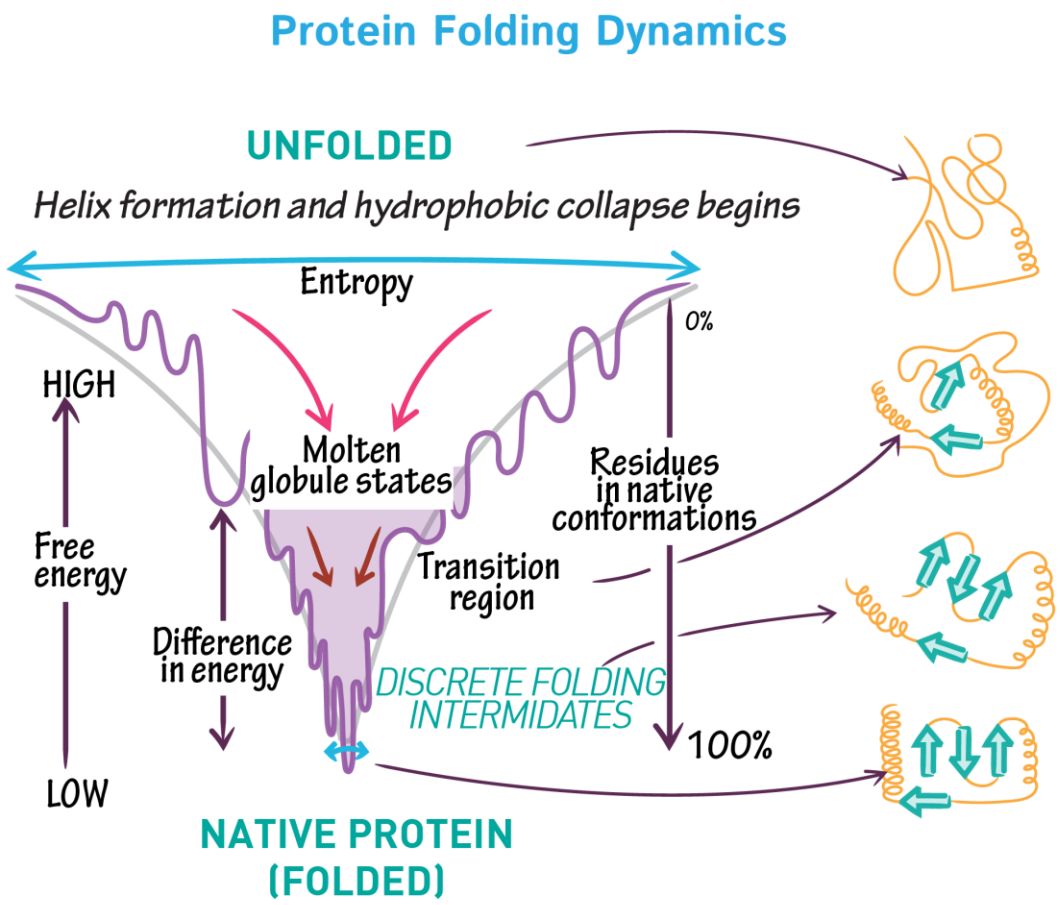
Loop modeling capabilities - Has sophisticated algorithms for modeling insertion/deletion regions (loops) that don't have template coverage, rather than just leaving gaps or using crude methods.

Optimization of side-chain packing - Uses molecular dynamics-based optimization with a specialized force field to refine side-chain conformations and relieve steric clashes.

Statistical potentials - Employs knowledge-based statistical potentials derived from known protein structures, capturing preferences for amino acid interactions beyond simple physics-based force fields.

Automated refinement - Performs conjugate gradient and molecular dynamics refinement to produce models with better geometry and fewer violations of physical constraints.

Ab initio modeling



Ab initio: (Latin) from the beginning.

Ab initio methods aim to predict protein structures from their amino acid sequences without relying on known structural templates.

These methods often use physics-based energy functions or machine learning techniques to sample the conformational space and identify low-energy structures.

These methods are very time-consuming since the configuration space is incredibly large.

Very interesting from a theoretical point of view, but do not produce reliable results yet.

ALPHAFOLD

AlphaFold

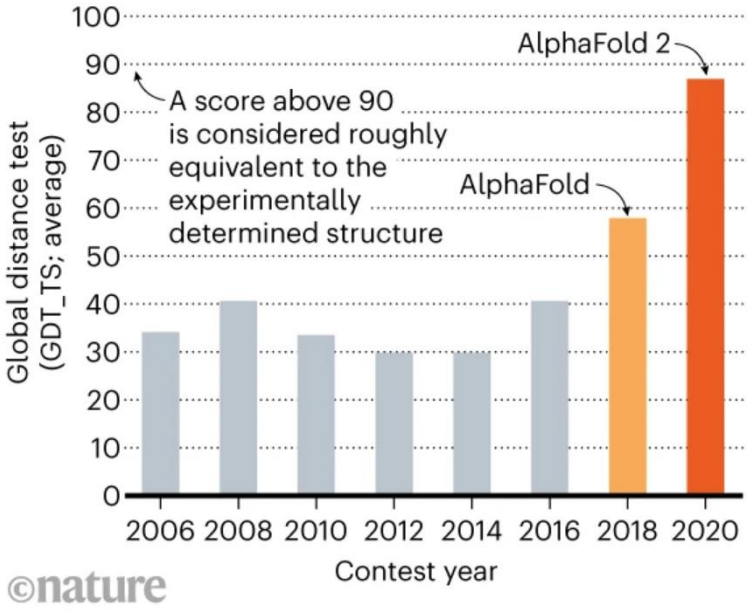
‘It will change everything’: DeepMind’s AI makes gigantic leap in solving protein structures

Google’s deep-learning program for determining the 3D shapes of proteins stands to transform biology, say scientists.

By Ewen Callaway

STRUCTURE SOLVER

DeepMind’s AlphaFold 2 algorithm significantly outperformed other teams at the CASP14 protein-folding contest — and its previous version’s performance at the last CASP.



AlphaFold utilizes deep learning and advanced machine learning techniques to accurately predict protein structures.

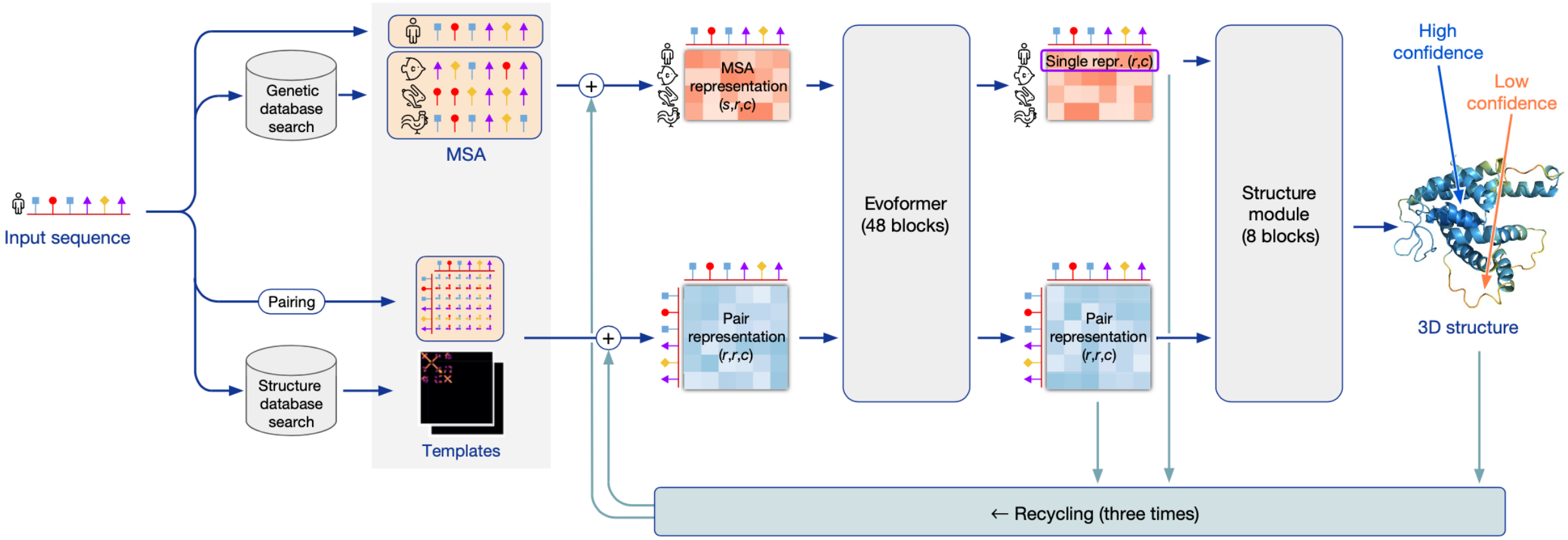
The key innovations of AlphaFold include:

1. **Attention-based Neural Network:** AlphaFold uses an attention-based neural network architecture that can capture long-range interactions between amino acids, which are crucial for accurate protein structure prediction.

2. **Evolutionary Information**: AlphaFold incorporates evolutionary information from multiple sequence alignments, which helps to improve the accuracy of its predictions.

3. **End-to-End Approach:** AlphaFold is an end-to-end approach that directly predicts the three-dimensional structure from the amino acid sequence without relying on intermediate steps or templates.

How does it work?

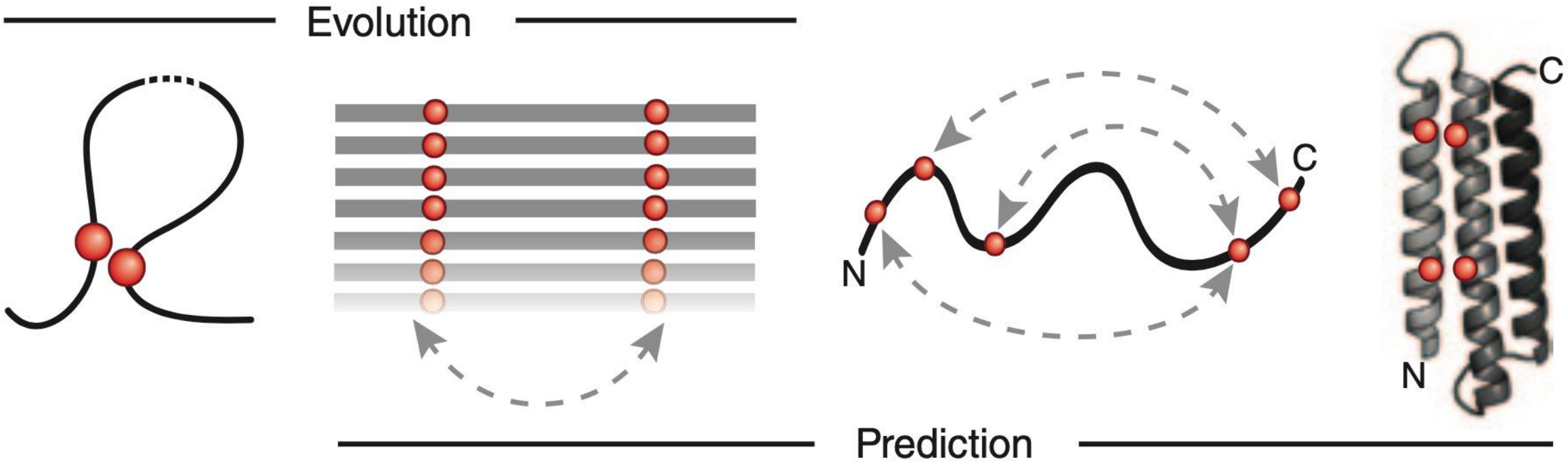


This and the next few slides are taken by a PPT from Kresten Lindorff-Larsen.

How does it work?

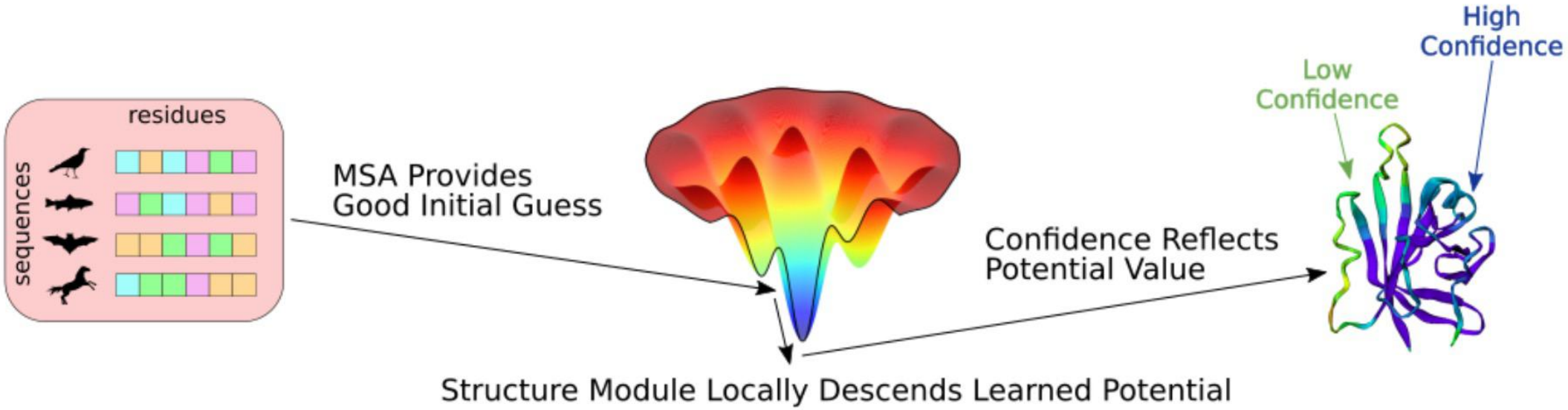
Slightly provocative takes:

- This is experimental data, and hence the AF2-derived structures are in some sense experimentally-derived
- Because the evolutionary data reports on the function of the proteins in cells, these models may in some cases represent the biologically relevant structures better than structures derived in vitro



Marks et al, Nature Biotech, 2012

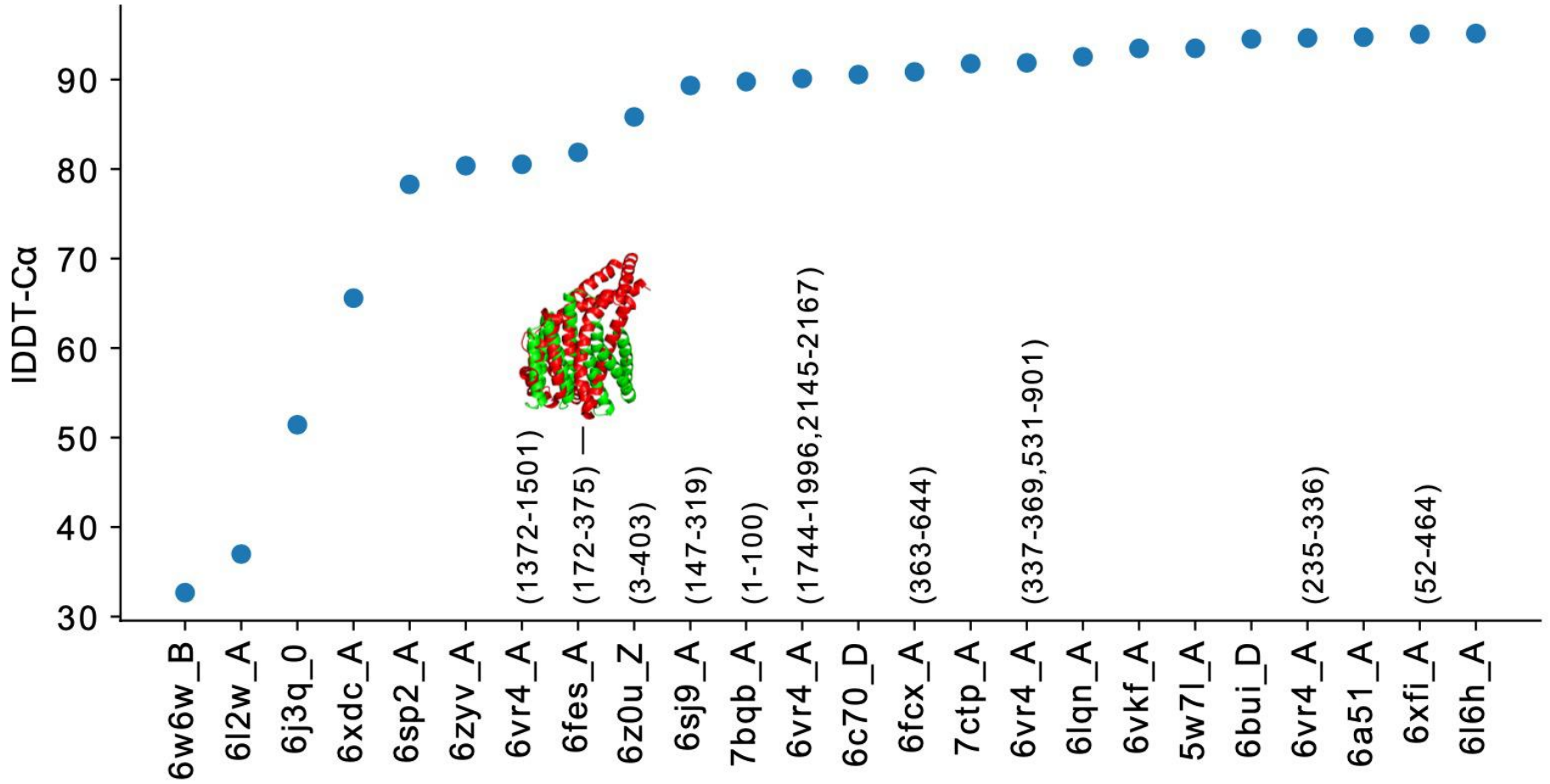
A guided search in a good energy function?



Roney and Ovchinnikov, bioRxiv, 2022

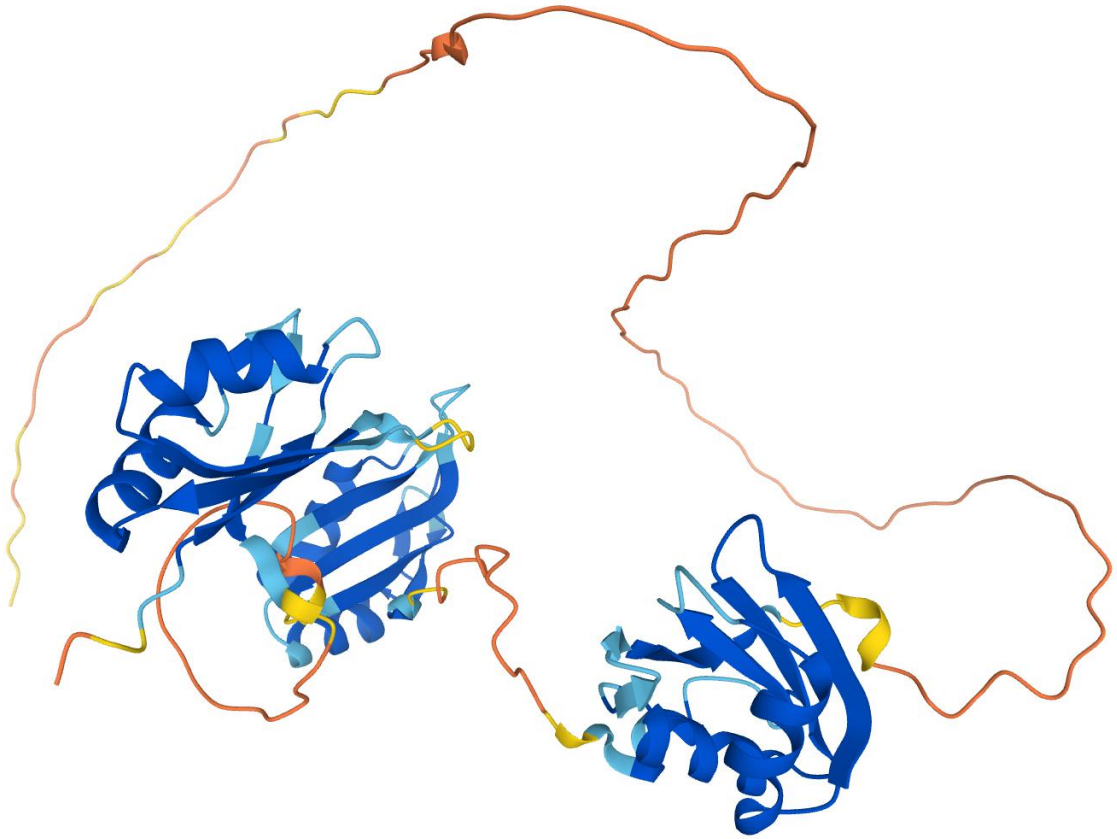
Is it just memorizing the PDB and/or doing homology modelling?

Performance on novel structures/folds (always a tricky thing to define) suggests not



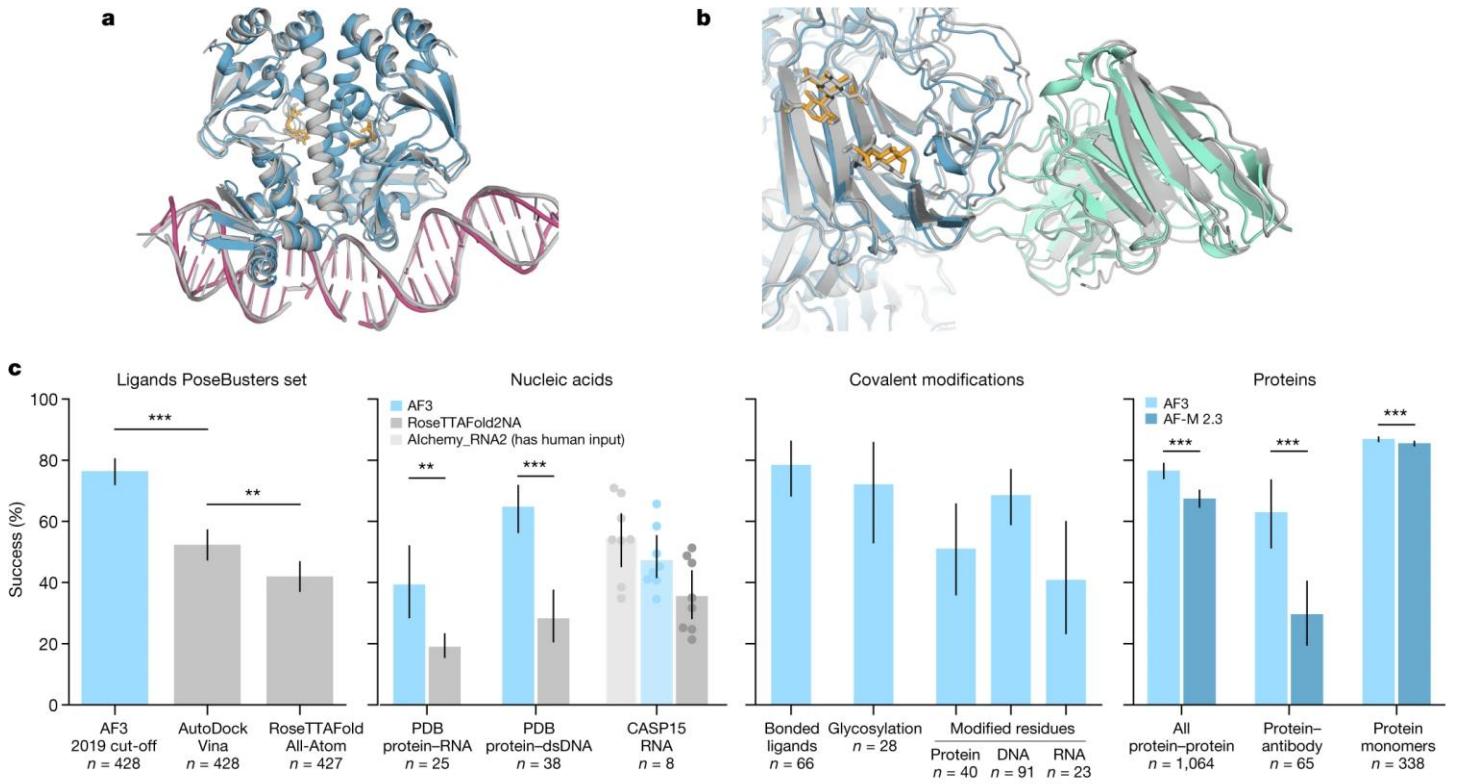
AF2 is a pretty good predictor of disorder

- AF2 was trained on monomeric proteins with structures resolved in the PDB
- It is not designed to predict flexibility or structures of flexible regions
- AF2, however, is pretty good at telling you when you should not trust the predictions
- When AF2 is unsure, the region is likely disordered*

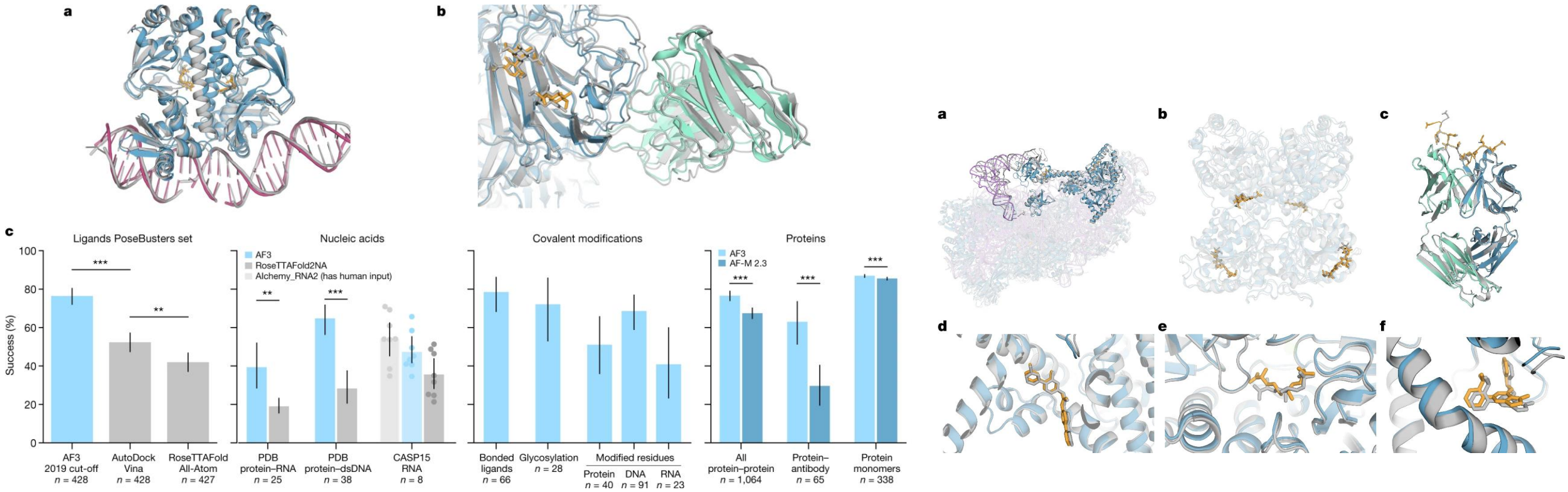


* Tunyasuvunakool et al, Nature, 2021
* Akdel et al, bioRxiv, 2021

AF3 accurately predicts structures across biomolecular complexes



AF3 accurately predicts structures across biomolecular complexes



Selected structure predictions from AF3. Predicted protein chains are shown in blue (predicted antibody in green), predicted ligands and glycans in orange, predicted RNA in purple and the ground truth is shown in grey. **a**, Human 40S small ribosomal subunit (7,663 residues) including 18S ribosomal RNA and Met-tRNA$_i^{Met}$ (opaque purple) in a complex with translation initiation factors eIF1A and eIF5B (opaque blue; PDB 7TQL; full-complex LDDT, 87.7; GDT, 86.9). **b**, The glycosylated globular portion of an EXTL3 homodimer (PDB 7AU2; mean pocket-aligned r.m.s.d., 1.10 Å). **c**, Mesothelin C-terminal peptide bound to the monoclonal antibody 15B6 (PDB 7U8C; DockQ, 0.85). **d**, LGK974, a clinical-stage inhibitor, bound to PORCN in a complex with the WNT3A peptide (PDB 7URD; ligand r.m.s.d., 1.00 Å). **e**, (5S,6S)-O7-sulfo DADH bound to the AziU3/U2 complex with a novel fold (PDB 7WUX; ligand r.m.s.d., 1.92 Å). **f**, Analogue of NIH-12848 bound to an allosteric site of PI5P4Kγ (PDB 7QIE; ligand r.m.s.d., 0.37 Å).

AlphaFold predictions on Uniprot

Function
Names & Taxonomy
Subcellular Location
Disease & Variants
PTM/Processing
Expression
Interaction
Structure
Family & Domains
Sequence
Similar Proteins

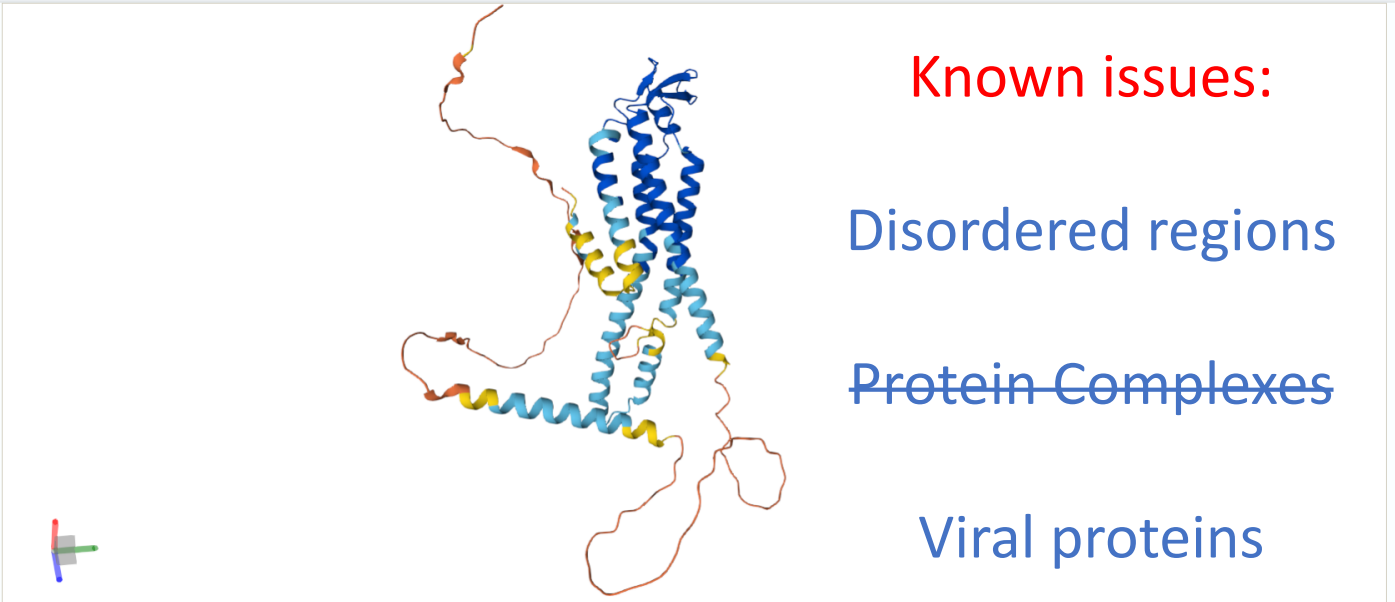
Entry | Variant viewer 437 | Feature viewer | Genomic coordinates | Publications | External links | History

Structure

Model Confidence:

- Very high (pLDDT > 90)
- Confident (90 > pLDDT > 70)
- Low (70 > pLDDT > 50)
- Very low (pLDDT < 50)

AlphaFold produces a per-residue confidence score (pLDDT) between 0 and 100. Some regions with low pLDDT may be unstructured in isolation.



Known issues:

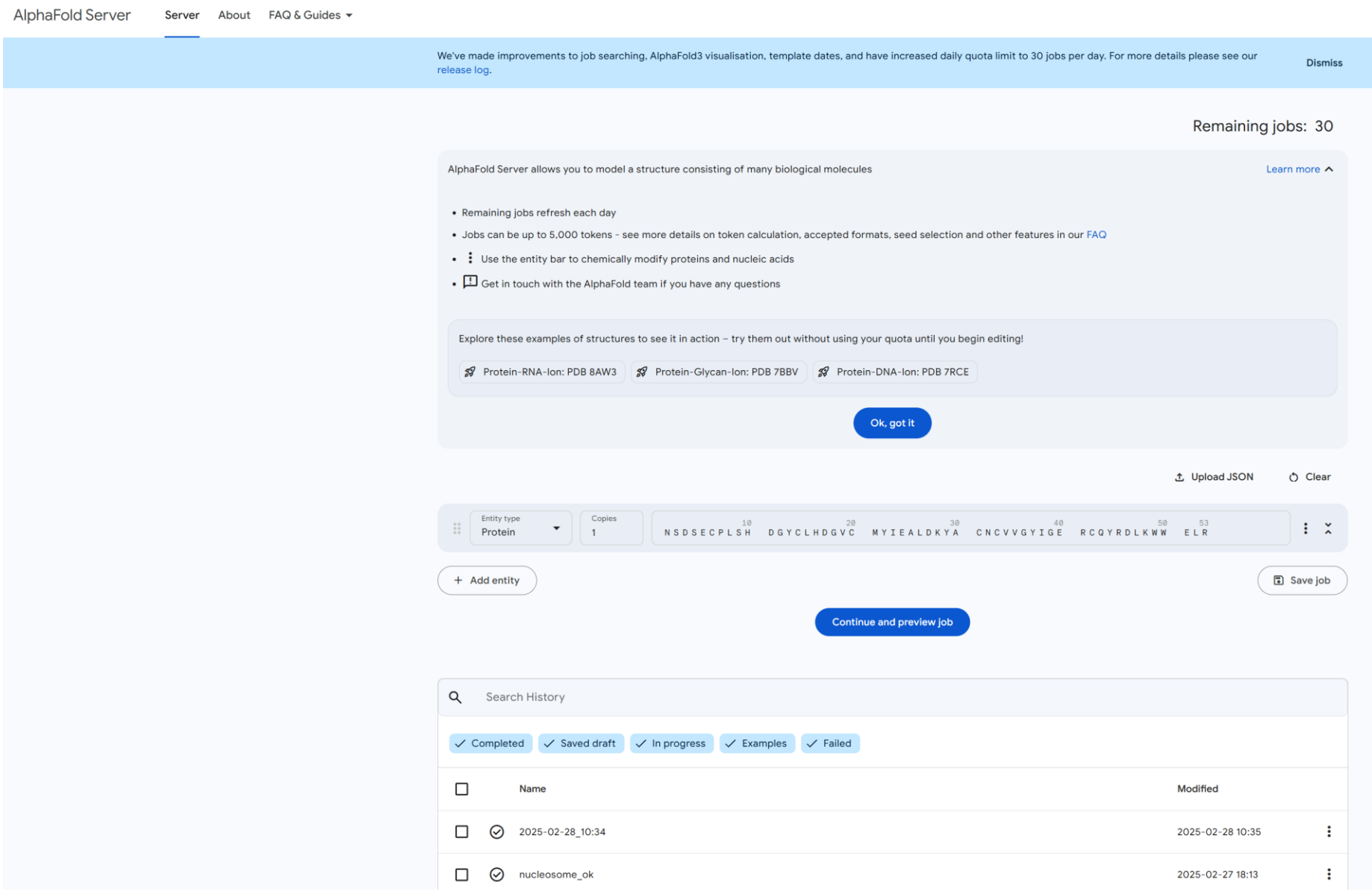
Disordered regions

~~Protein Complexes~~

Viral proteins

SOURCE	IDENTIFIER	METHOD	RESOLUTION	CHAIN	POSITIONS	LINKS
-- Select --		-- Select --				
PDB	7XQD	EM	2.70 Å	A/B/C/D/E/F/G/H/I/J/K/L	1-257	PDBe · RCSB-PDB · PDBj · PDBsum · Foldseek
PDB	7XQF	EM	2.30 Å	A/B/C/D/E/F/G/H/I/J/K/L	1-257	PDBe · RCSB-PDB · PDBj · PDBsum · Foldseek
PDB	7XQG	EM	3.80 Å	A/B/C/D/E/F	1-257	PDBe · RCSB-PDB · PDBj · PDBsum · Foldseek
PDB	7XQH	EM	3.80 Å	A/B/C/D/E/F	1-257	PDBe · RCSB-PDB · PDBj · PDBsum · Foldseek
PDB	7XQI	EM	3.70 Å	A/B/C/D/E/F	1-257	PDBe · RCSB-PDB · PDBj · PDBsum · Foldseek
PDB	7XQJ	EM	4.00 Å	A/B/C/D/E/F	1-257	PDBe · RCSB-PDB · PDBj · PDBsum · Foldseek
PDB	7Z1T	EM	2.26 Å	A/B/C/D/E/F/G/H/I/J/K/L	1-382	PDBe · RCSB-PDB · PDBj · PDBsum · Foldseek
PDB	7Z22	EM	2.95 Å	A/B/C/D/E/F/G/H/I/J/K/L	1-382	PDBe · RCSB-PDB · PDBj · PDBsum · Foldseek
PDB	7Z23	EM	3.98 Å	A/B/C/D/E/F	1-382	PDBe · RCSB-PDB · PDBj · PDBsum · Foldseek
AlphaFold	AF-P17302-F1	Predicted			1-382	AlphaFold · Foldseek

AlphaFold 3


AlphaFold Server
Server
About
FAQ & Guides
We've made improvements to job searching, AlphaFold3 visualisation, template dates, and have increased daily quota limit to 30 jobs per day. For more details please see our release log.
Dismiss
Remaining jobs: 30
AlphaFold Server allows you to model a structure consisting of many biological molecules
Learn more
Remaining jobs refresh each day
Jobs can be up to 5,000 tokens - see more details on token calculation, accepted formats, seed selection and other features in our FAQ
Use the entity bar to chemically modify proteins and nucleic acids
Get in touch with the AlphaFold team if you have any questions
Explore these examples of structures to see it in action – try them out without using your quota until you begin editing!
Protein-RNA-Ion: PDB 8AW3
Protein-Glycan-Ion: PDB 7BBV
Protein-DNA-Ion: PDB 7RCE
Ok, got it
Upload JSON
Clear
Entity type
Protein
Copies
1
NSDSECPLSH DGYCLHDGVC MYIEALDKYA CNCVVGYIGE RCQYRDLKWW ELR
Add entity
Save job
Continue and preview job
Search History
Completed
Saved draft
In progress
Examples
Failed
Name
Modified
2025-02-28_10:34
2025-02-28 10:35
nucleosome_ok
2025-02-27 18:13